



# Cluster Analysis II

## Lecture 12

肖升生 博士

Dr. Shengsheng Xiao

Email: [Xiao.shengsheng@shufe.edu.cn](mailto:Xiao.shengsheng@shufe.edu.cn)



# Outline

---

- Hierarchical Clustering
- DBSACN
- Cluster Validation
- Case Study



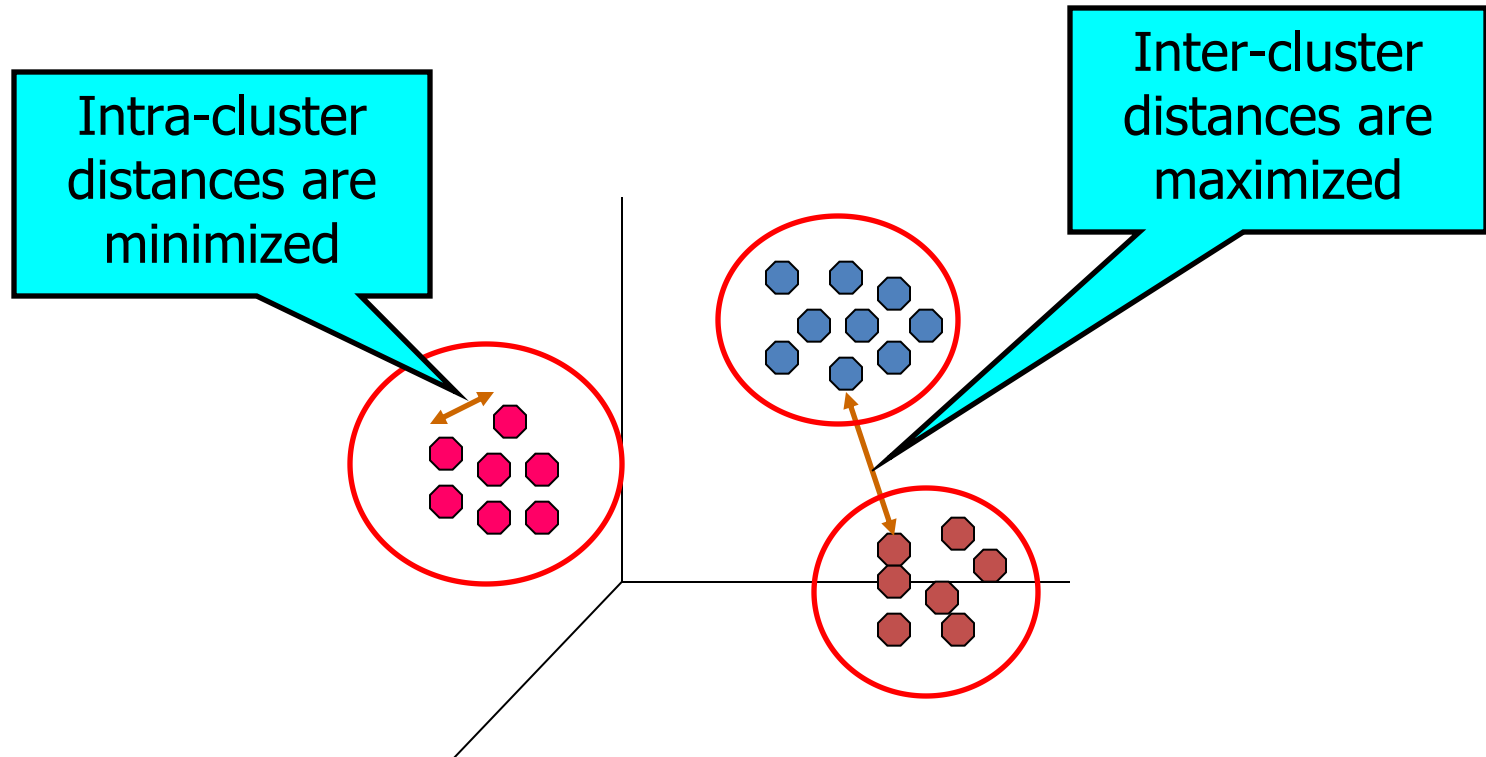
# Outline

---

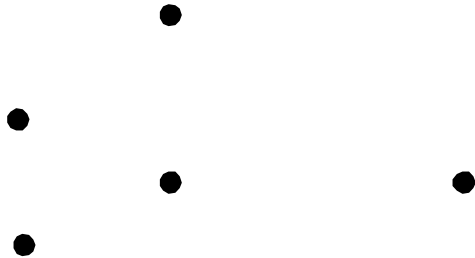
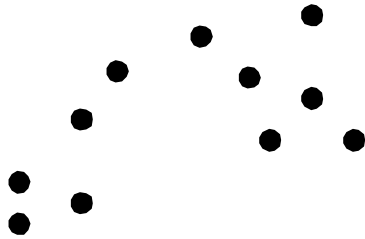
- Hierarchical Clustering
- DBSACN
- Cluster Validation
- Case Study

# What is Cluster Analysis?

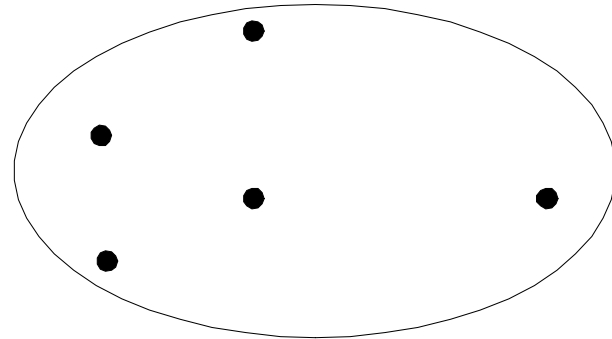
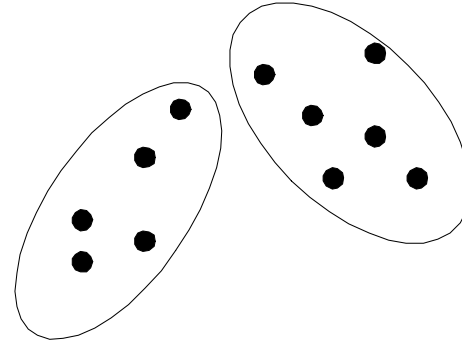
- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



# Partitional Clustering

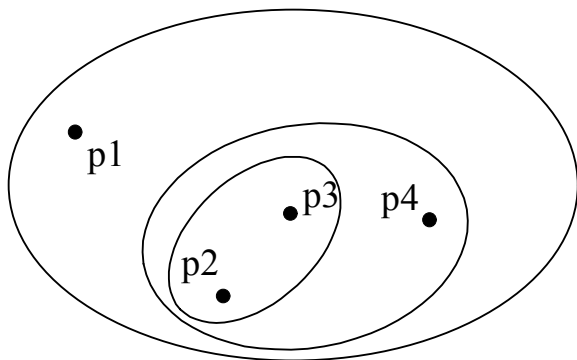


Original Points

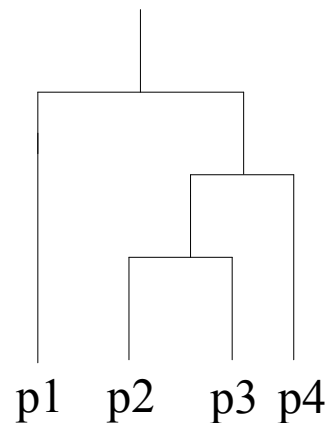


A Partitional Clustering

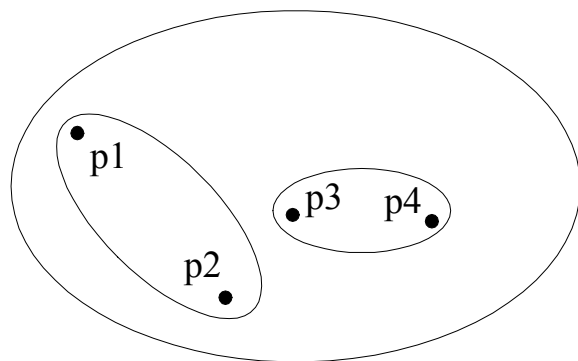
# Hierarchical Clustering



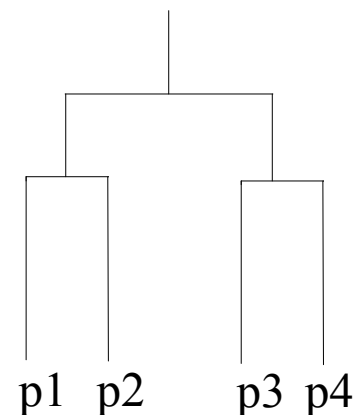
Traditional Hierarchical Clustering



Traditional Dendrogram



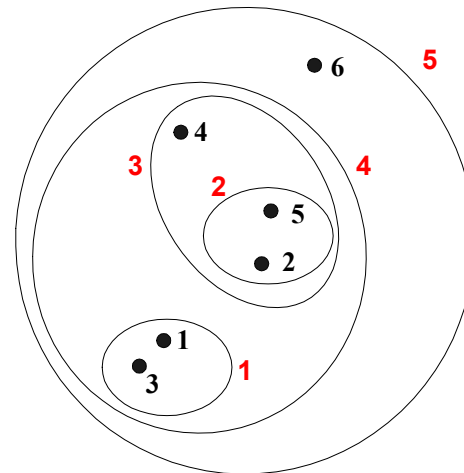
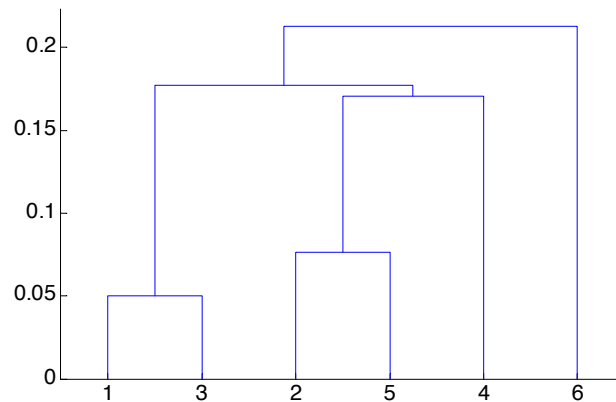
Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

# Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
  - A tree like diagram that records the sequences of merges or splits



# Strengths of Hierarchical Clustering

---

- Do not have to assume any particular number of clusters
  - Any desired number of clusters can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful taxonomies
  - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)



# Hierarchical Clustering

---

- Two main types of hierarchical clustering
  - Agglomerative:
    - Start with the points as individual clusters
    - At each step, merge the closest pair of clusters until only one cluster (or  $k$  clusters) left
  - Divisive:
    - Start with one, all-inclusive cluster
    - At each step, split a cluster until each cluster contains a point (or there are  $k$  clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
  - Merge or split one cluster at a time

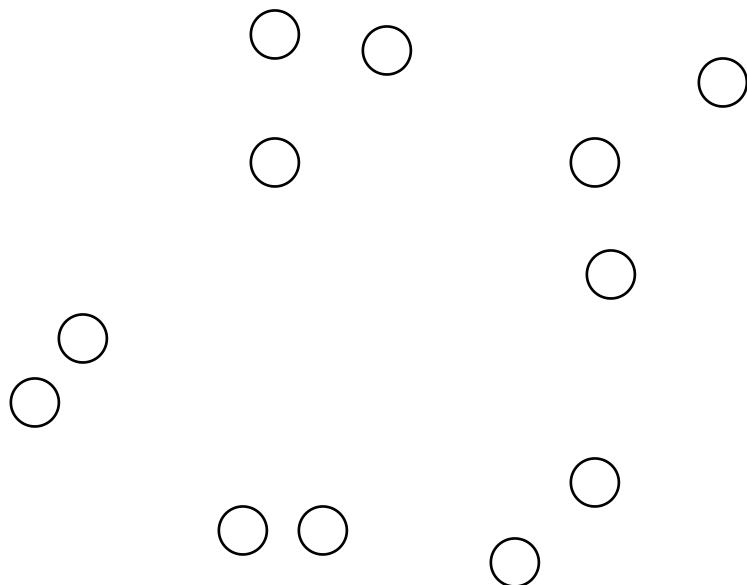
# Agglomerative Clustering Algorithm

---

- Basic algorithm is straightforward
  1. Compute the proximity matrix
  2. Let each data point be a cluster
  3. **Repeat**
  4. Merge the two closest clusters
  5. Update the proximity matrix
  6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
  - Different approaches to defining the distance between clusters distinguish the different algorithms

# Starting Situation

- Start with clusters of individual points and a proximity matrix



|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix

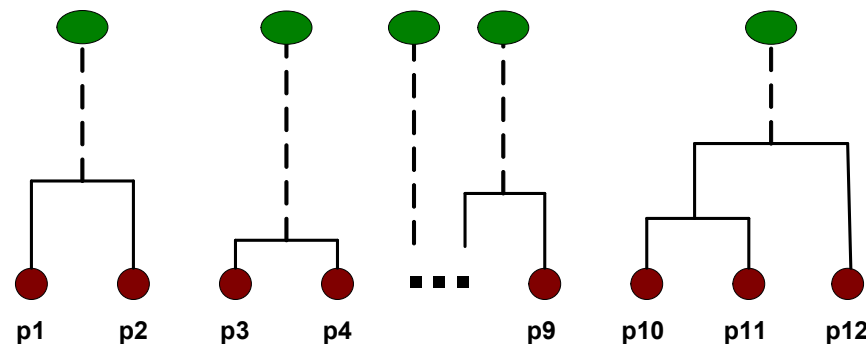
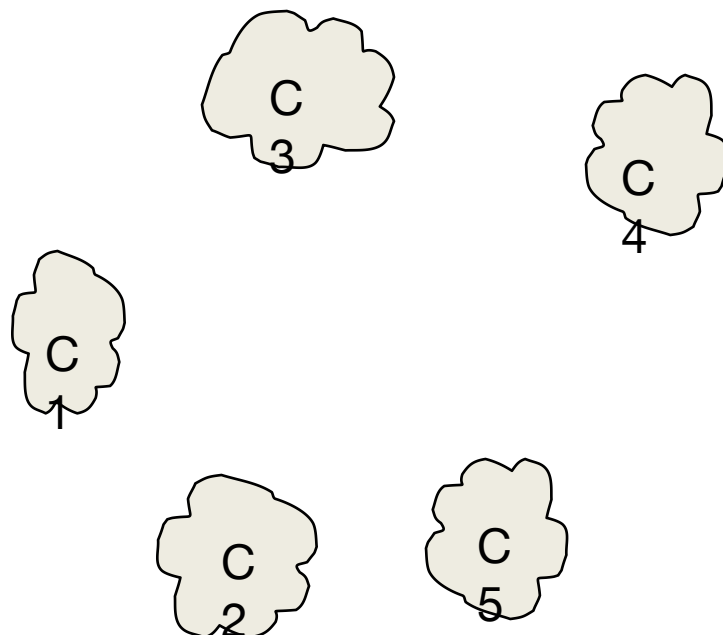


# Intermediate Situation

- After some merging steps, we have some clusters

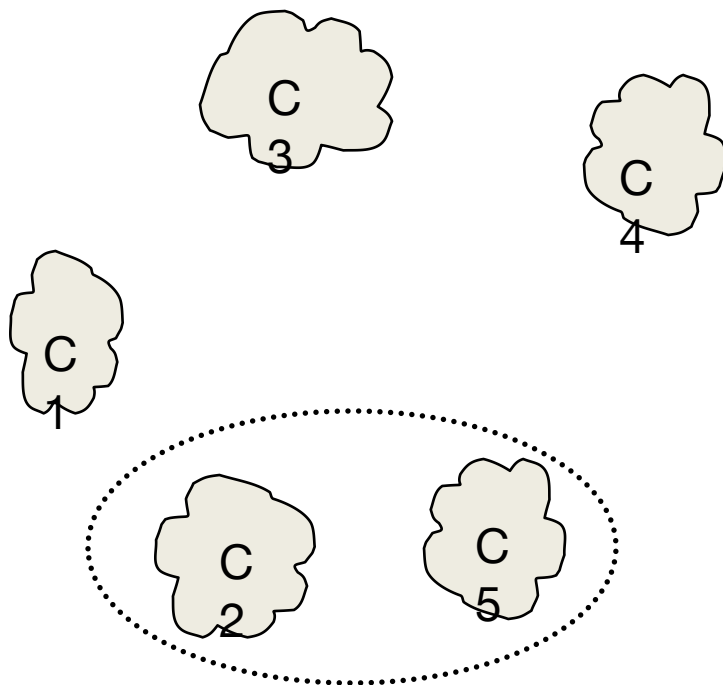
|    | C1 | C2 | C3 | C4 | C5 |
|----|----|----|----|----|----|
| C1 |    |    |    |    |    |
| C2 |    |    |    |    |    |
| C3 |    |    |    |    |    |
| C4 |    |    |    |    |    |
| C5 |    |    |    |    |    |

Proximity Matrix



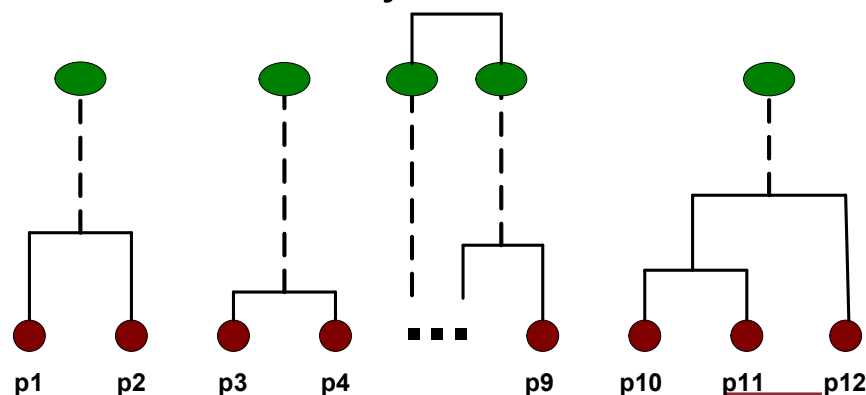
# Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



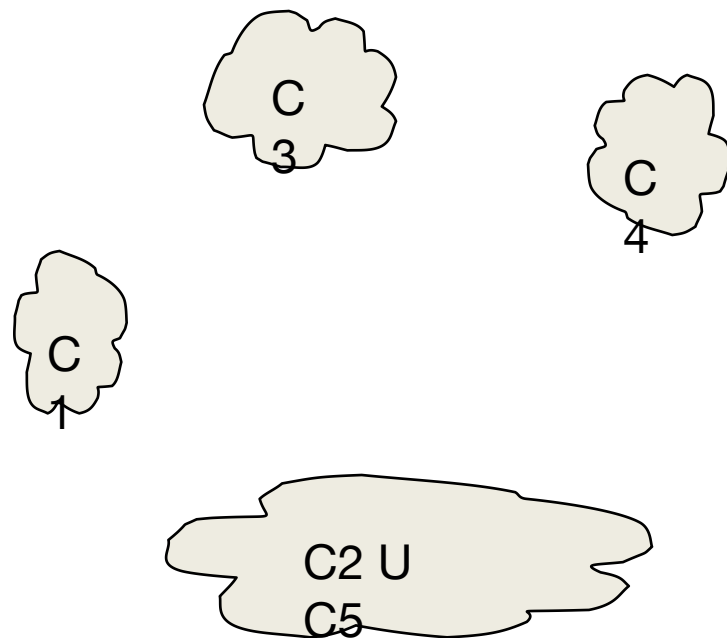
|    | C1 | C2 | C3 | C4 | C5 |
|----|----|----|----|----|----|
| C1 |    |    |    |    |    |
| C2 |    |    |    |    |    |
| C3 |    |    |    |    |    |
| C4 |    |    |    |    |    |
| C5 |    |    |    |    |    |

Proximity Matrix



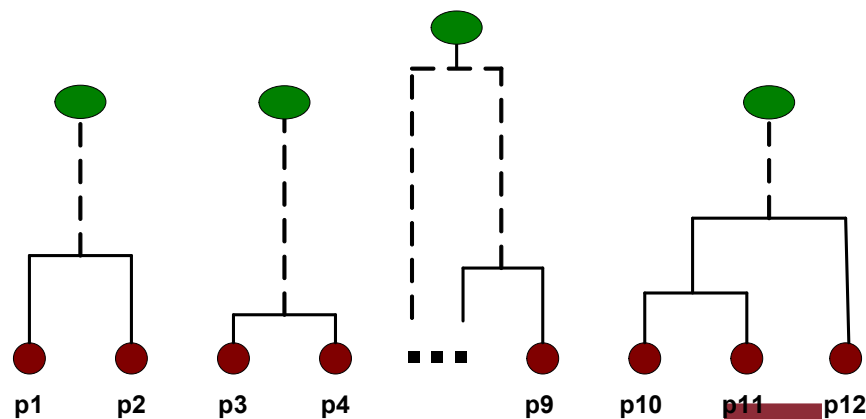
# After Merging

- The question is “How do we update the proximity matrix?”

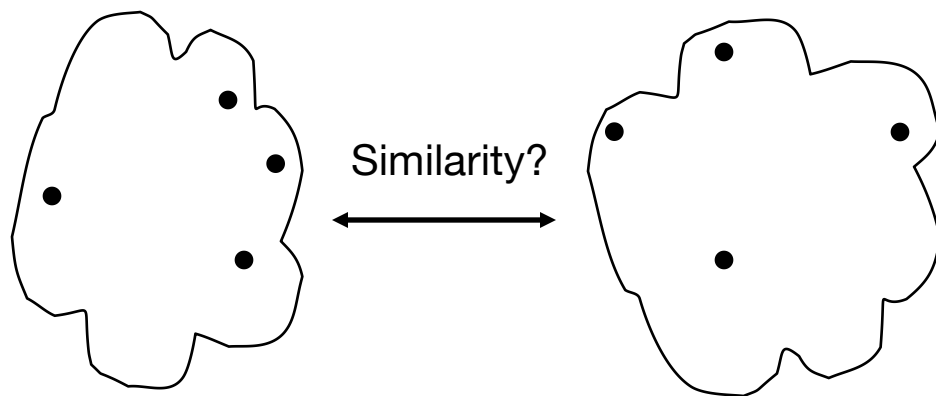


|            |  | C1 | C2<br>U<br>C5 | C3 | C4 |   |
|------------|--|----|---------------|----|----|---|
| C1         |  |    | ?             |    |    |   |
| C2 U<br>C5 |  | ?  | ?             |    | ?  | ? |
| C3         |  |    | ?             |    |    |   |
| C4         |  |    | ?             |    |    |   |

Proximity Matrix



# How to Define Inter-Cluster Similarity

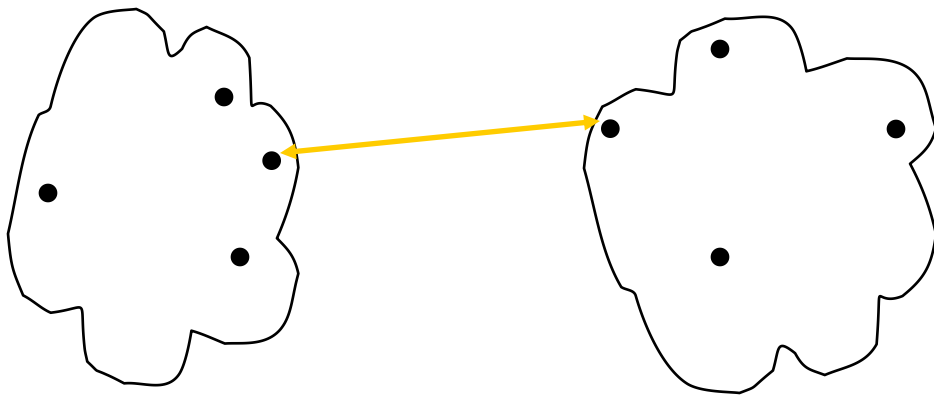


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix

# How to Define Inter-Cluster Similarity



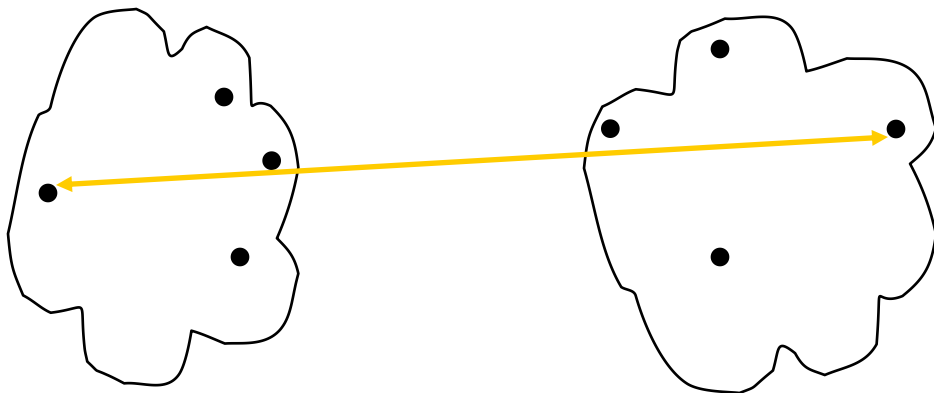
- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix



# How to Define Inter-Cluster Similarity

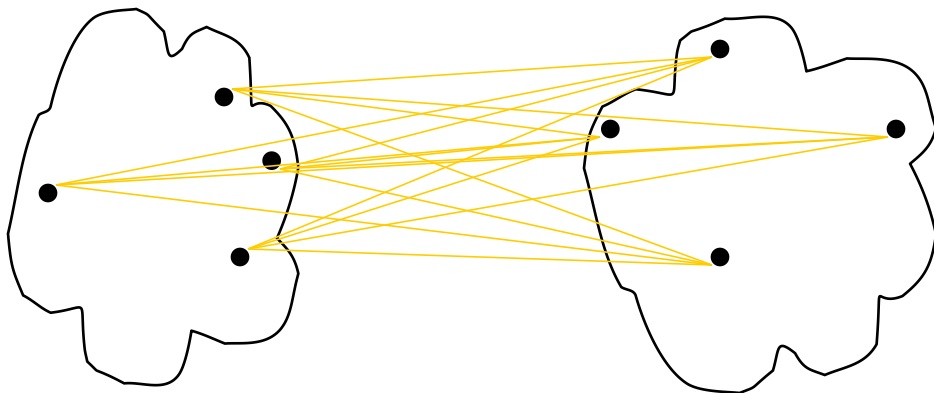


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix

# How to Define Inter-Cluster Similarity

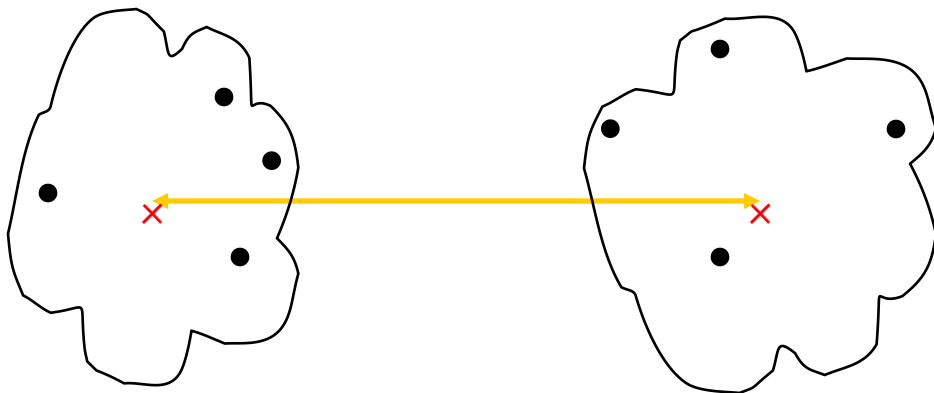


- MIN
- MAX
- **Group Average**
- Distance Between Centroids
- Other methods driven by an objective function
  - Ward's Method uses squared error

|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix

# How to Define Inter-Cluster Similarity



- MIN
- MAX
- Group Average
- **Distance Between Centroids**
- Other methods driven by an objective function
  - Ward's Method uses squared error

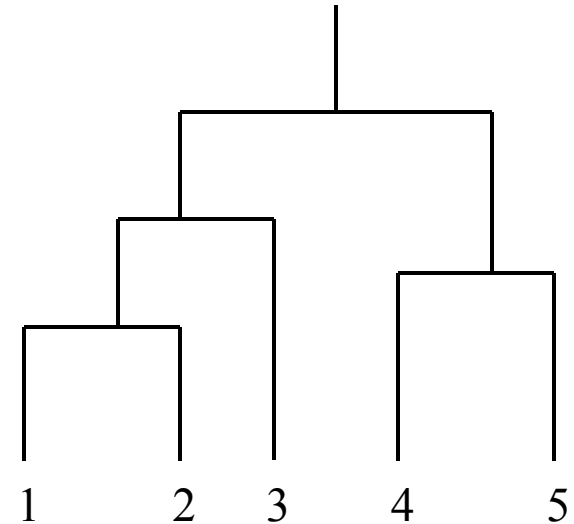
|    | p1 | p2 | p3 | p4 | p5 | ... |
|----|----|----|----|----|----|-----|
| p1 |    |    |    |    |    |     |
| p2 |    |    |    |    |    |     |
| p3 |    |    |    |    |    |     |
| p4 |    |    |    |    |    |     |
| p5 |    |    |    |    |    |     |
| .  |    |    |    |    |    |     |

Proximity Matrix

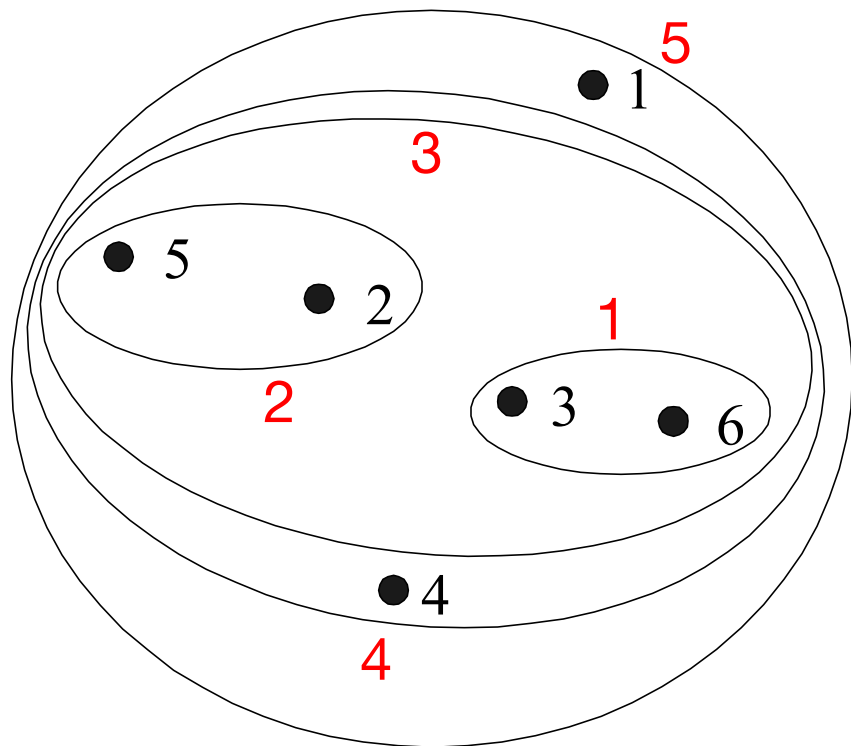
# Cluster Similarity: MIN or Single Link

- Similarity of two clusters is based on the two most similar (closest) points in the different clusters
  - Determined by one pair of points, i.e., by one link in the proximity graph.

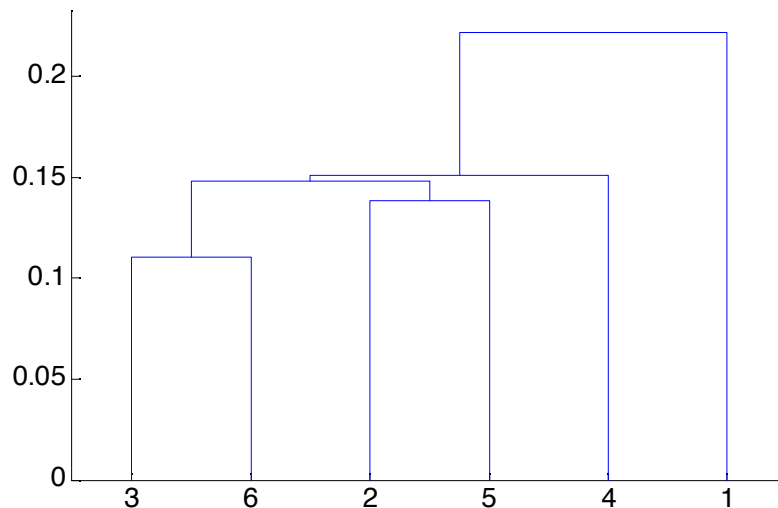
|    | I1   | I2   | I3   | I4   | I5   |
|----|------|------|------|------|------|
| I1 | 1.00 | 0.90 | 0.10 | 0.65 | 0.20 |
| I2 | 0.90 | 1.00 | 0.70 | 0.60 | 0.50 |
| I3 | 0.10 | 0.70 | 1.00 | 0.40 | 0.30 |
| I4 | 0.65 | 0.60 | 0.40 | 1.00 | 0.80 |
| I5 | 0.20 | 0.50 | 0.30 | 0.80 | 1.00 |



# Hierarchical Clustering: MIN



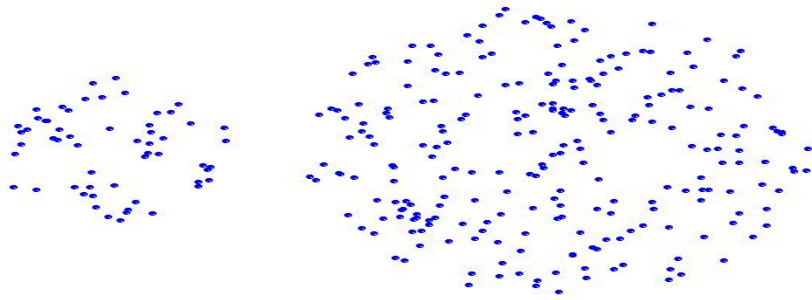
Nested Clusters



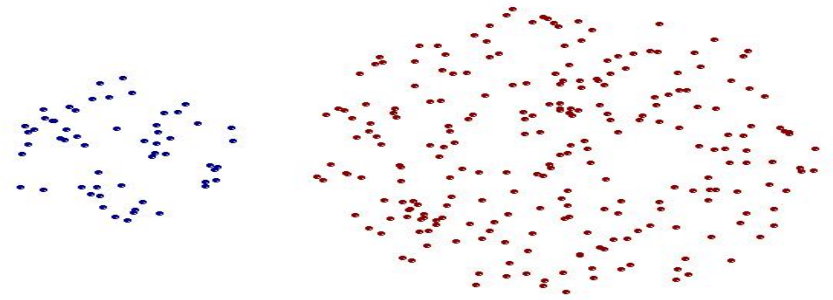
Dendrogram

# Strength of MIN

---



Original Points

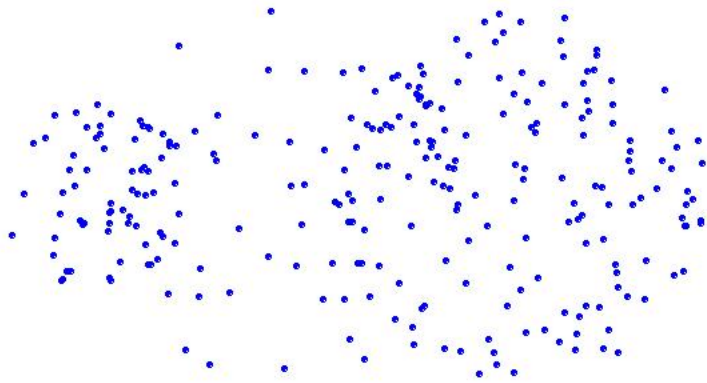


Two Clusters

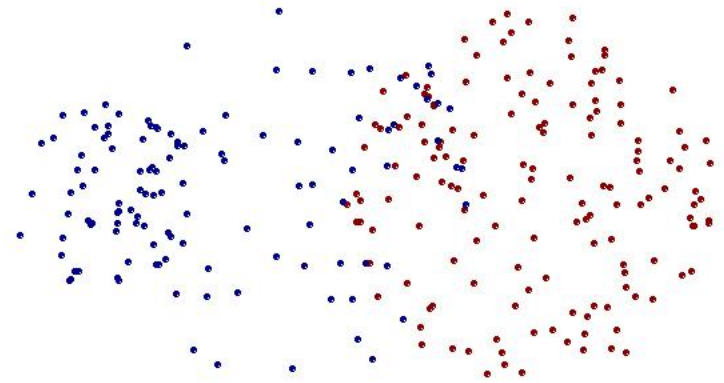
- Can handle non-elliptical shapes

# Limitations of MIN

---



Original Points



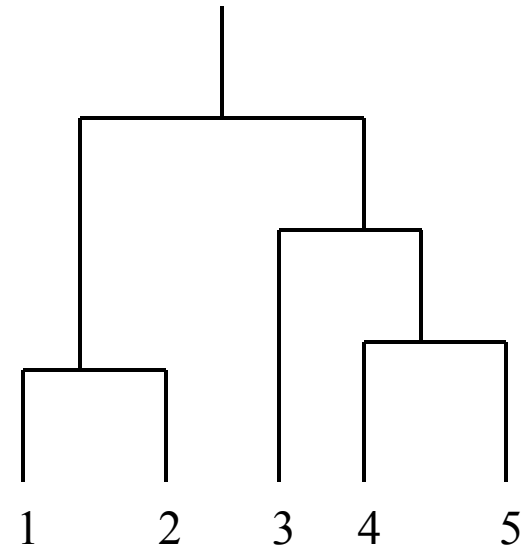
Two Clusters

- Sensitive to noise and outliers

# Cluster Similarity: MAX or Complete Linkage

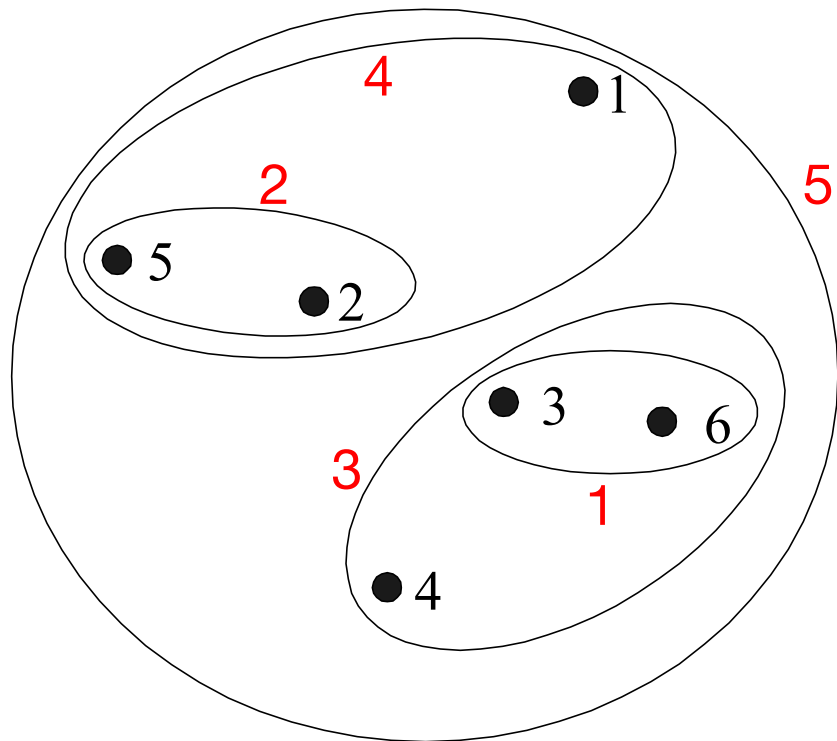
- Similarity of two clusters is based on the two least similar (most distant) points in the different clusters
  - Determined by all pairs of points in the two clusters

|    | I1   | I2   | I3   | I4   | I5   |
|----|------|------|------|------|------|
| I1 | 1.00 | 0.90 | 0.10 | 0.65 | 0.20 |
| I2 | 0.90 | 1.00 | 0.70 | 0.60 | 0.50 |
| I3 | 0.10 | 0.70 | 1.00 | 0.40 | 0.30 |
| I4 | 0.65 | 0.60 | 0.40 | 1.00 | 0.80 |
| I5 | 0.20 | 0.50 | 0.30 | 0.80 | 1.00 |

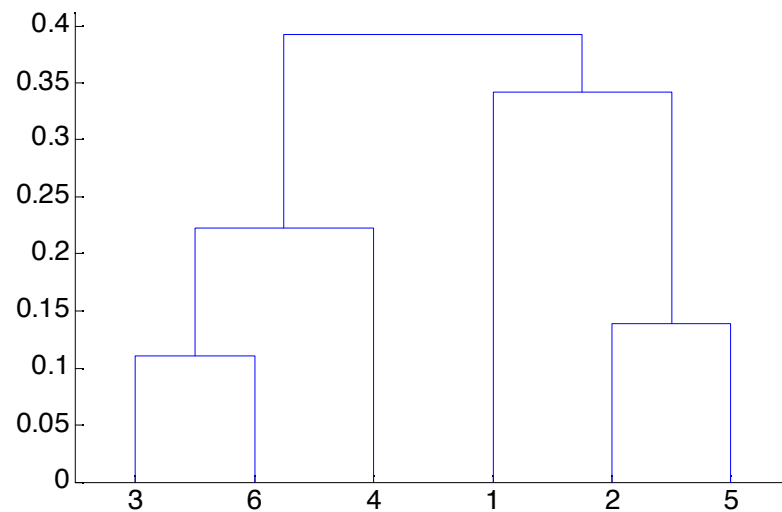




# Hierarchical Clustering: MAX



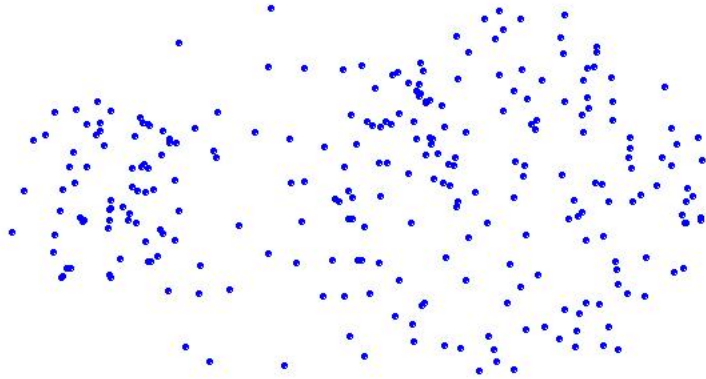
Nested Clusters



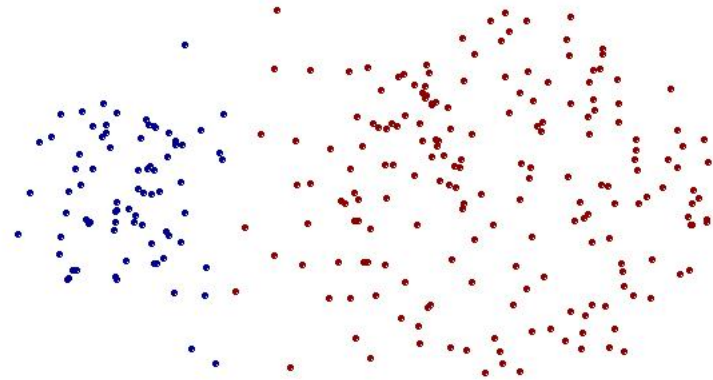
Dendrogram

# Strength of MAX

---



Original Points

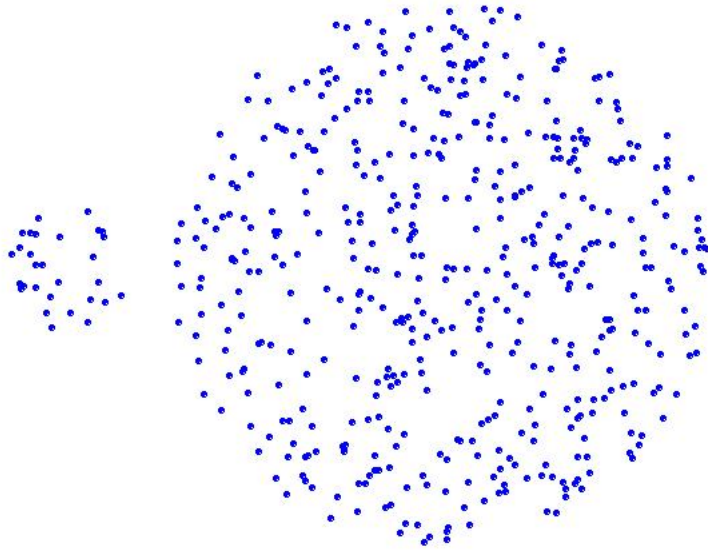


Two Clusters

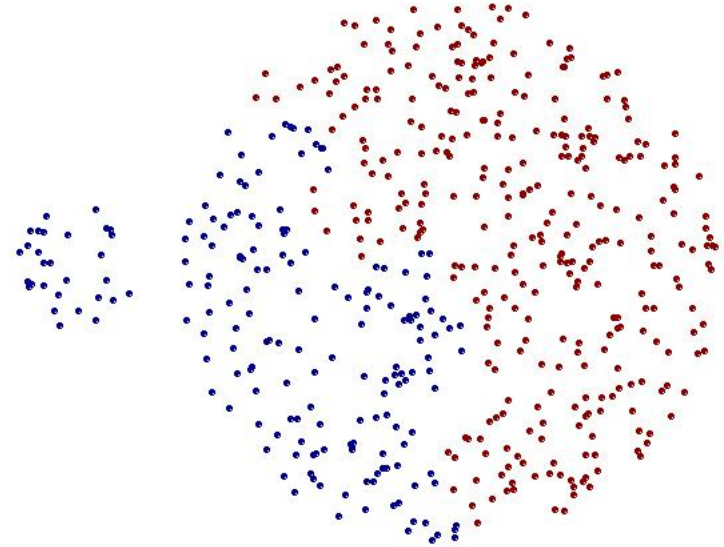
- Less susceptible to noise and outliers

# Limitations of MAX

---



Original Points



Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

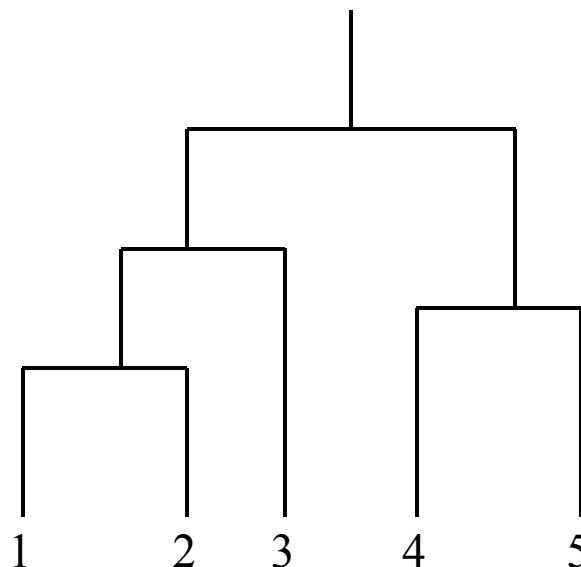
# Cluster Similarity: Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

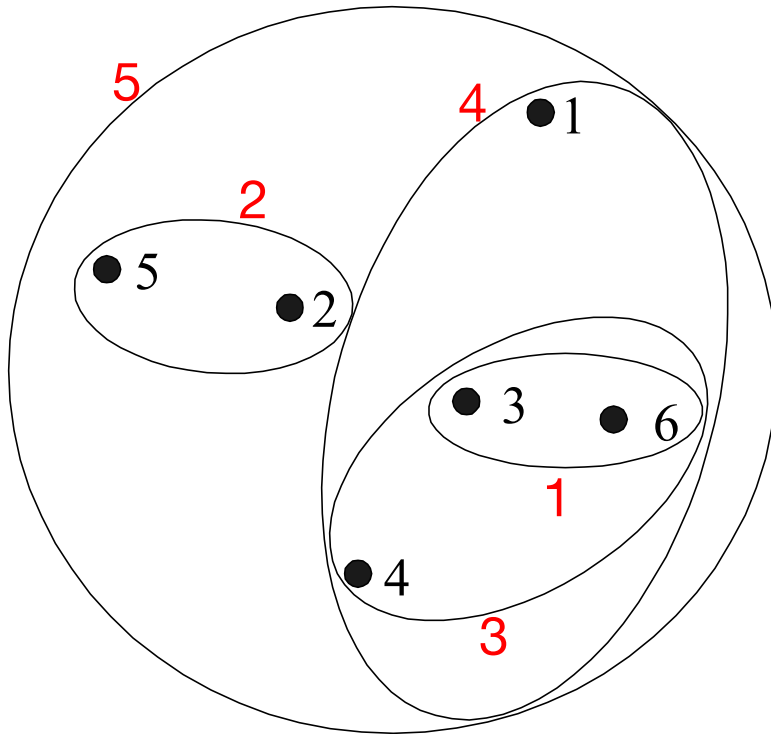
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| * |\text{Cluster}_j|}$$

- Need to use average connectivity for scalability since total proximity favors large clusters

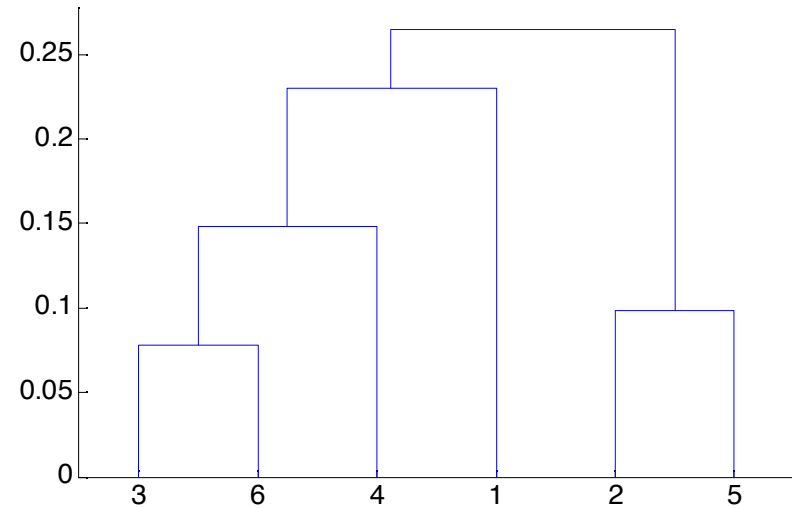
|    | I1   | I2   | I3   | I4   | I5   |
|----|------|------|------|------|------|
| I1 | 1.00 | 0.90 | 0.10 | 0.65 | 0.20 |
| I2 | 0.90 | 1.00 | 0.70 | 0.60 | 0.50 |
| I3 | 0.10 | 0.70 | 1.00 | 0.40 | 0.30 |
| I4 | 0.65 | 0.60 | 0.40 | 1.00 | 0.80 |
| I5 | 0.20 | 0.50 | 0.30 | 0.80 | 1.00 |



# Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram



# Hierarchical Clustering: Group Average

---

- Compromise between Single and Complete Link
- Strengths
  - Less susceptible to noise and outliers
- Limitations
  - Biased towards globular clusters



# Cluster Similarity: Ward's Method

---

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
  - Can be used to initialize K-means

# Hierarchical Clustering: Limitations

---

- Once a decision is made to combine two clusters, it cannot be undone
- No objective function is directly minimized
- Different schemes have problems with one or more of the following:
  - Sensitivity to noise and outliers
  - Difficulty handling different sized clusters and convex shapes
  - Breaking large clusters





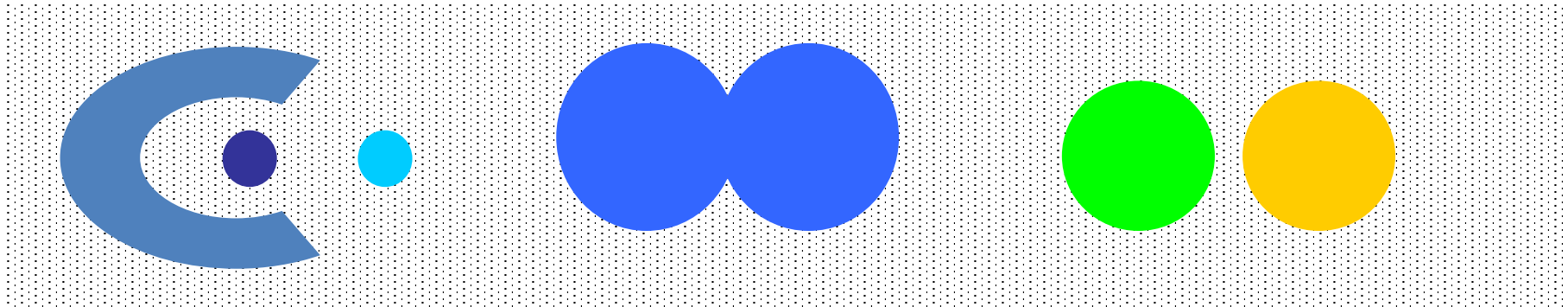
# Outline

---

- Hierarchical Clustering
- DBSACN
- Cluster Validation
- Case Study

# Density-Based Clusters

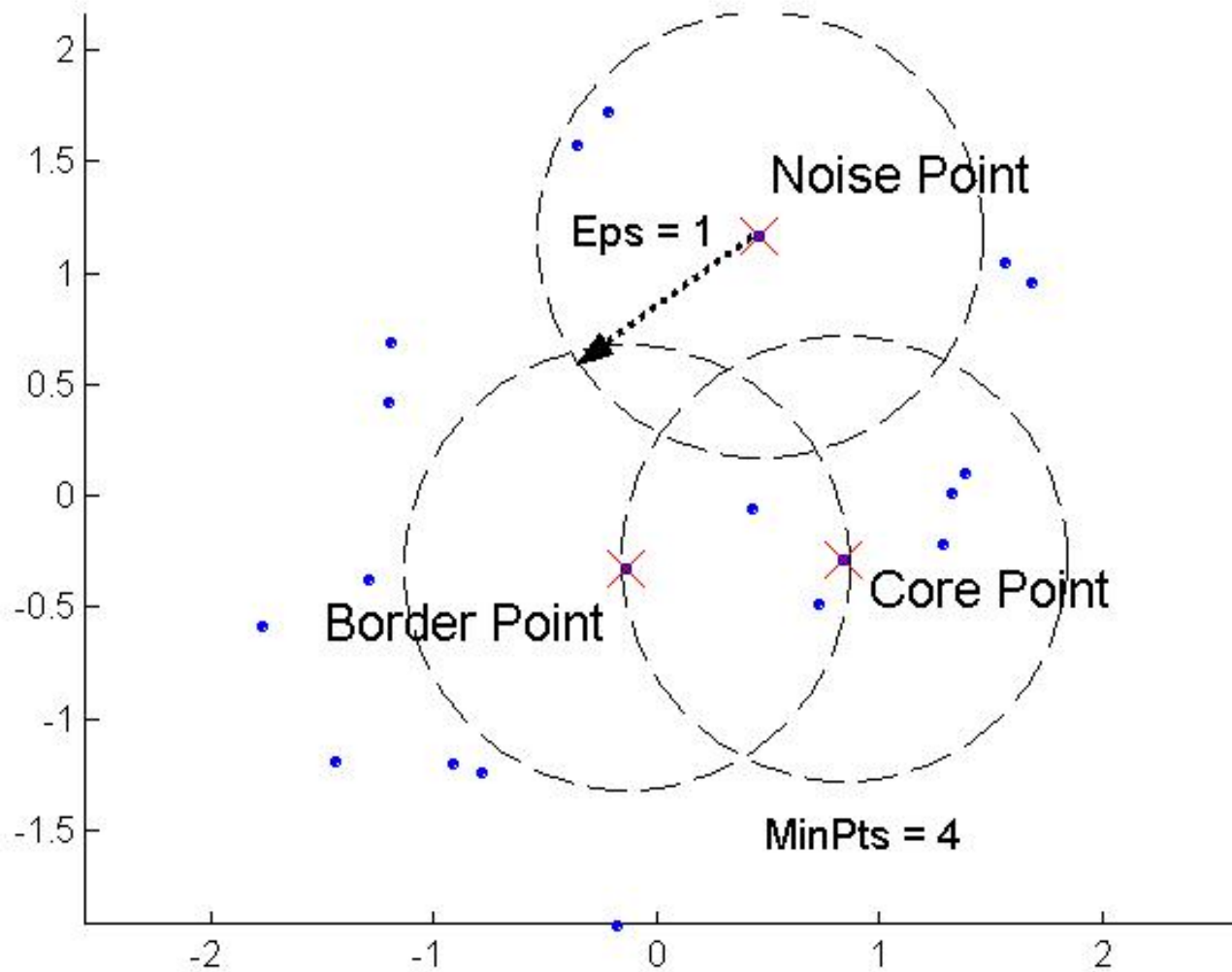
- Density-based Clusters
  - A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
  - Used when the clusters are irregular or intertwined, and when noise and outliers are present.



6 density-based clusters

- DBSCAN is a density-based algorithm.
  - Density = number of points within a specified radius (Eps)
  - A point is a **core point** if it has more than a specified number of points (MinPts) within Eps
    - These are points that are at the interior of a cluster
  - A **border point** has fewer than MinPts within Eps, but is in the neighborhood of a core point
  - A **noise point** is any point that is not a core point or a border point.

# DBSCAN: Core, Border, and Noise Points



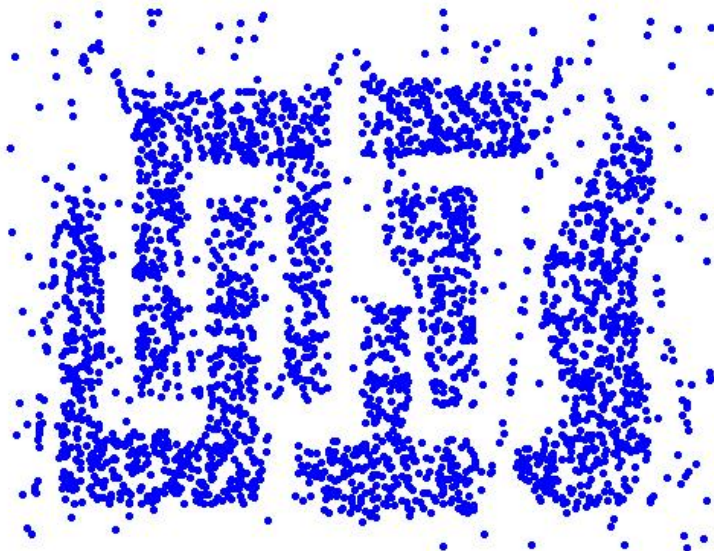
# DBSCAN Algorithm

---

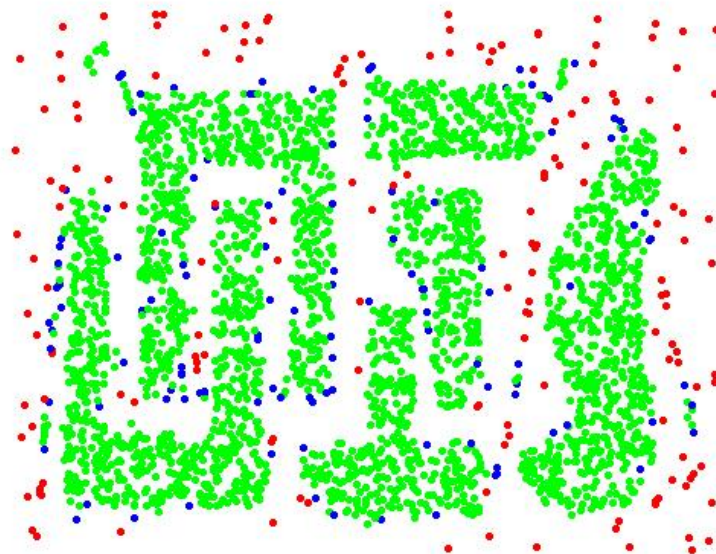
- Eliminate noise points
- Perform clustering on the remaining points

```
current_cluster_label  $\leftarrow$  1
for all core points do
  if the core point has no cluster label then
    current_cluster_label  $\leftarrow$  current_cluster_label + 1
    Label the current core point with cluster label current_cluster_label
  end if
  for all points in the Eps-neighborhood, except  $i^{th}$  the point itself do
    if the point does not have a cluster label then
      Label the point with cluster label current_cluster_label
    end if
  end for
end for
```

# DBSCAN: Core, Border and Noise Points



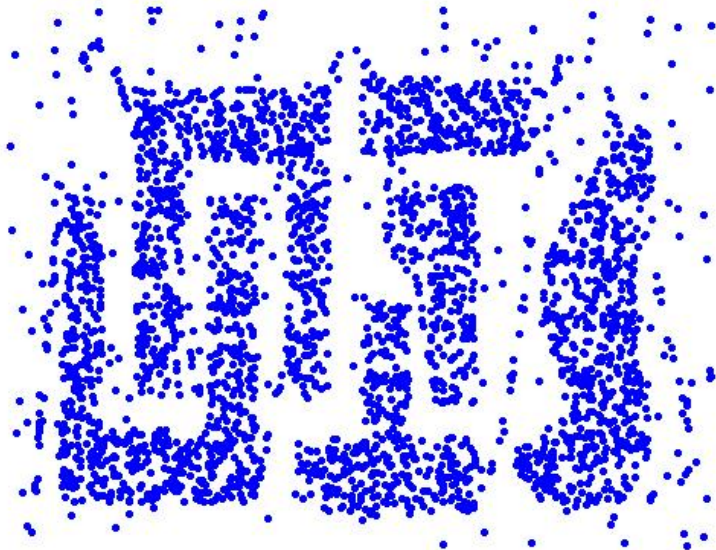
Original Points



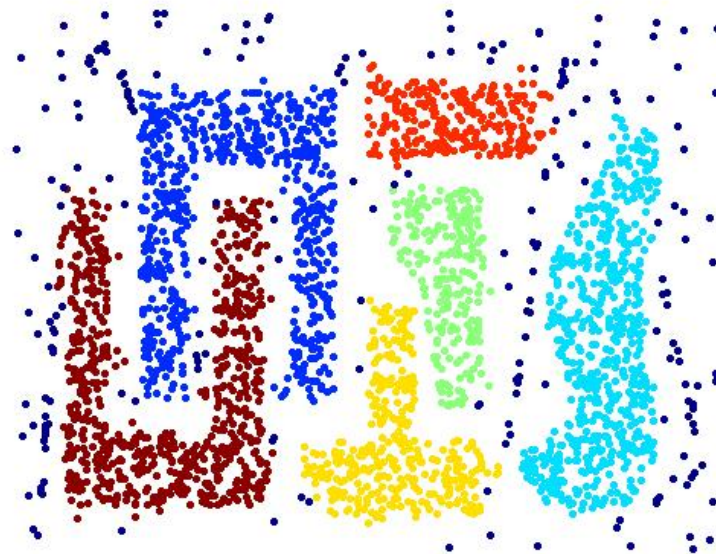
Point types: **core**,  
**border** and **noise**

Eps = 10, MinPts = 4

# When DBSCAN Works Well



Original Points

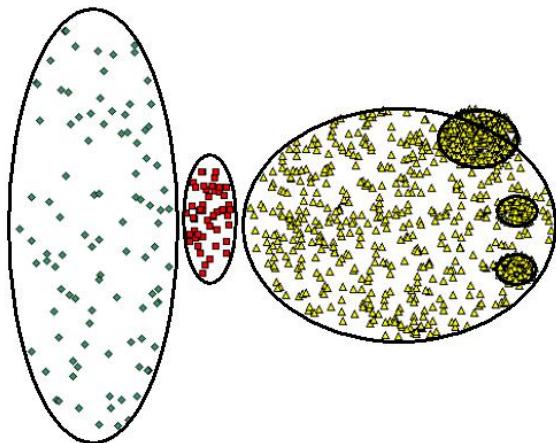


Clusters

- Resistant to Noise
- Can handle clusters of different shapes and sizes

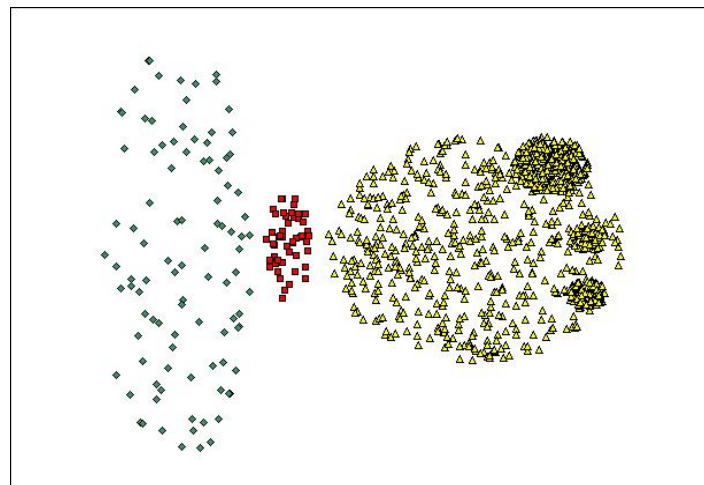


# When DBSCAN Does NOT Work Well

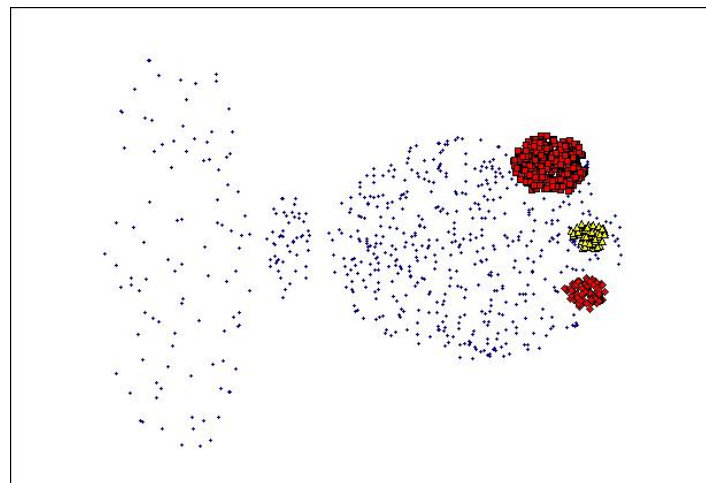


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.75).

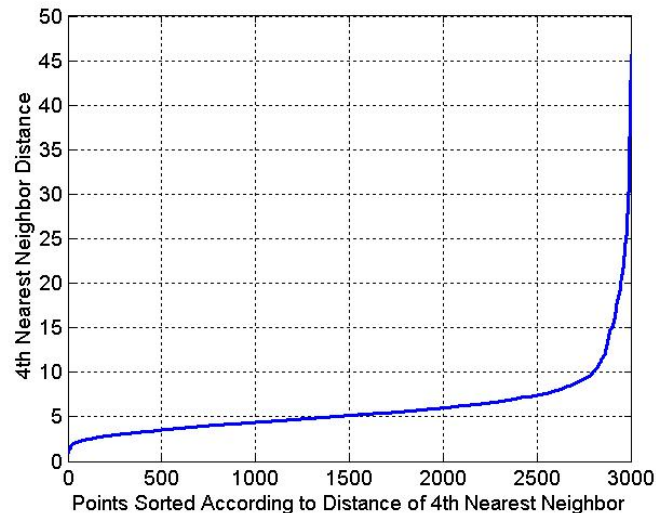


(MinPts=4, Eps=9.92)



# DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their  $k^{\text{th}}$  nearest neighbors are at roughly the same distance
- Noise points have the  $k^{\text{th}}$  nearest neighbor at farther distance
- So, plot sorted distance of every point to its  $k^{\text{th}}$  nearest neighbor





# Outline

---

- Hierarchical Clustering
- DBSACN
- Cluster Validation
- Case Study

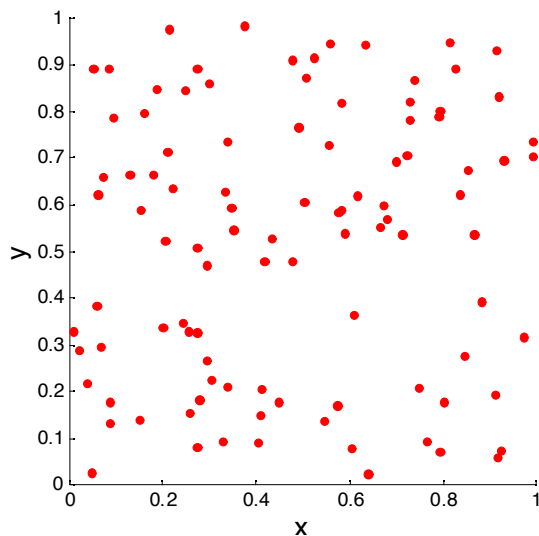
# Cluster Validity

---

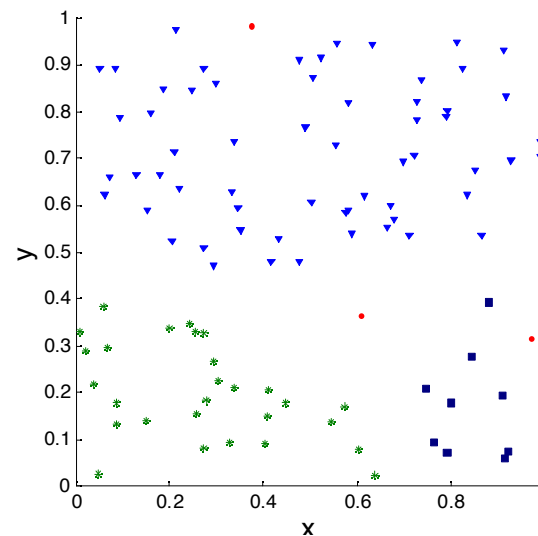
- For supervised classification we have a variety of measures to evaluate how good our model is
  - Accuracy, precision, recall
- For cluster analysis, the analogous question is **how to evaluate the “goodness” of the resulting clusters?**
- But “clusters are in the eye of the beholder”!
- why do we want to evaluate them?
  - To avoid finding patterns in noise
  - To compare clustering algorithms
  - To compare two sets of clusters
  - To compare two clusters

# Clusters found in Random Data

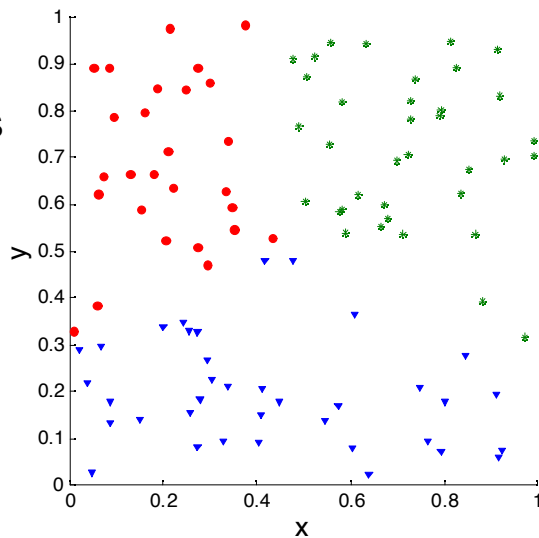
Random Points



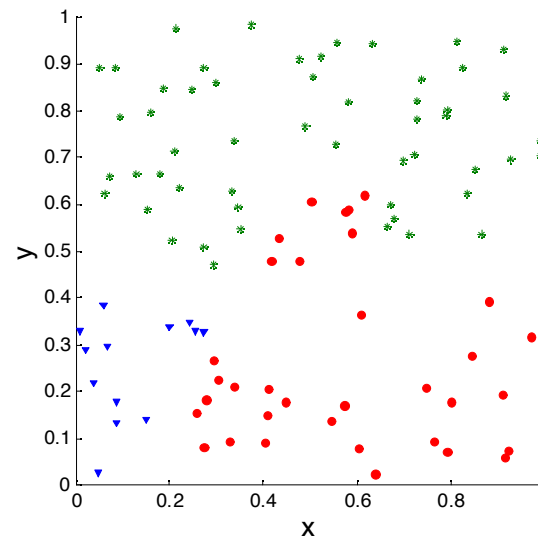
DBSCAN



K-means



Complete Link



# Measures of Cluster Validity

---

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
  - **Unsupervised Methods: internal Index** used to measure the goodness of a clustering structure *without* respect to external information.
    - Sum of Squared Error (SSE)
  - **Supervised Methods: external Index** used to measure the extent to which cluster labels match externally supplied class labels.
    - Entropy
  - **Between: Relative Index** used to compare two different clusterings or clusters.
    - Often an external or internal index is used for this function, e.g., SSE or entropy

# Internal Index: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
  - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
  - Cohesion is measured by the within cluster sum of squares (SSE)

$$SSE = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

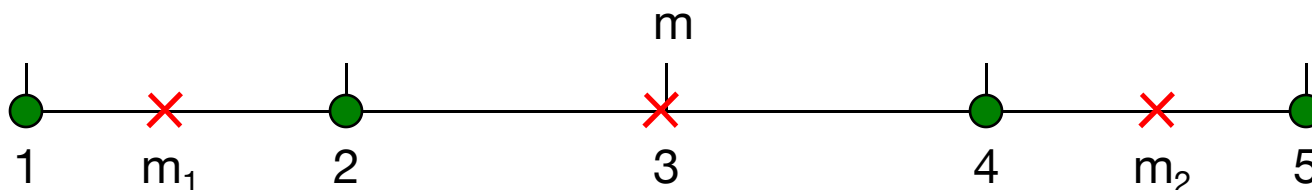
- Separation is measured by the between cluster sum of squares

$$SSB = \sum_i |C_i| (m - m_i)^2$$

- Where  $|C_i|$  is the size of cluster  $i$

# Internal Index: Cohesion and Separation

- Example: SSE
  - $SSE + SSB = SST = \text{constant}$



K=1 cluster:

$$SSE = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$SSB = 4 \times (3 - 3)^2 = 0$$

$$SST = 10 + 0 = 10$$

K=2 clusters:

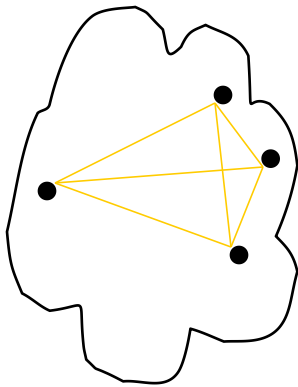
$$SSE = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$SSB = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

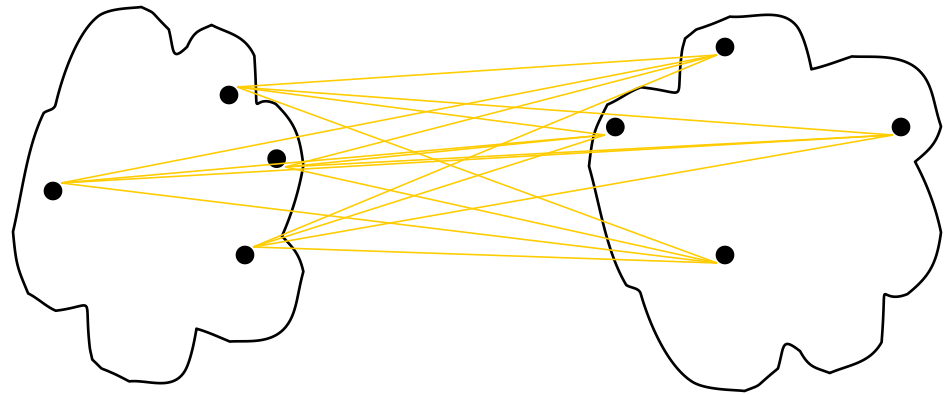
$$SST = 1 + 9 = 10$$

# Internal Index: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
  - Cluster cohesion is the sum of the weight of all links within a cluster.
  - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion



separation

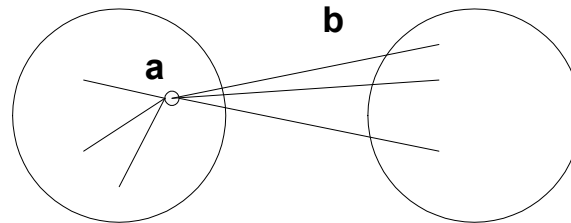


# Internal Index: Silhouette Coefficient

- Silhouette Coefficient (轮廓系数) combine ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point,  $i$ 
  - Calculate  $a$  = average distance of  $i$  to the points in its cluster
  - Calculate  $b$  = min (average distance of  $i$  to points in another cluster)
  - The silhouette coefficient for a point is then given by :

$$s_i = (b-a)/\max(a, b)$$

- Typically between -1 and 1.
- The closer to 1 the better.



- Can calculate the Average Silhouette width for a cluster or a clustering

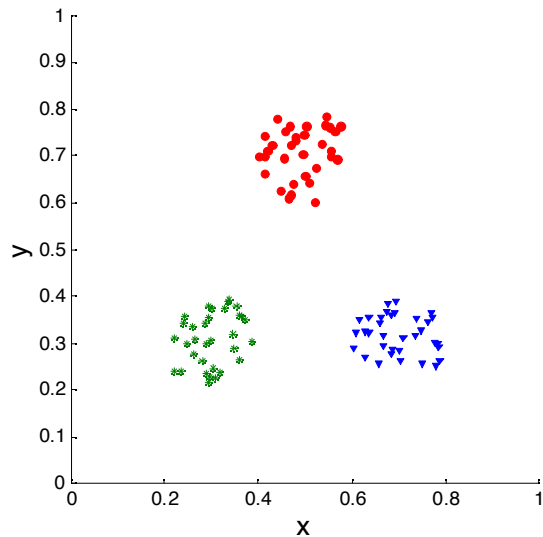
# Internal Index and Proximity Matrix

---

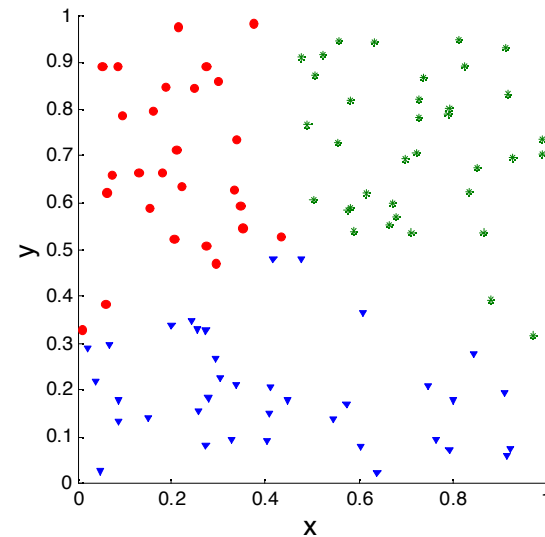
- Two matrices
  - Proximity Matrix
  - “Incidence” Matrix (关联矩阵)
    - One row and one column for each data point
    - An entry is 1 if the associated pair of points belong to the same cluster
    - An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
  - Since the matrices are symmetric, only the correlation between  $n(n-1) / 2$  entries needs to be calculated.
- High correlation indicates that points that belong to the same cluster are close to each other.
- Not a good measure for some density or contiguity based clusters.

# Cluster Validity Via Correlation

- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



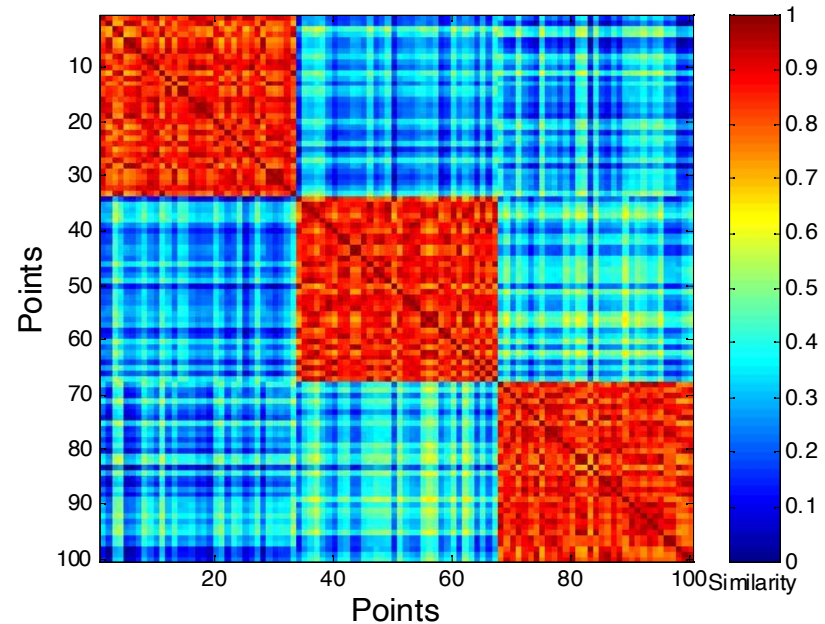
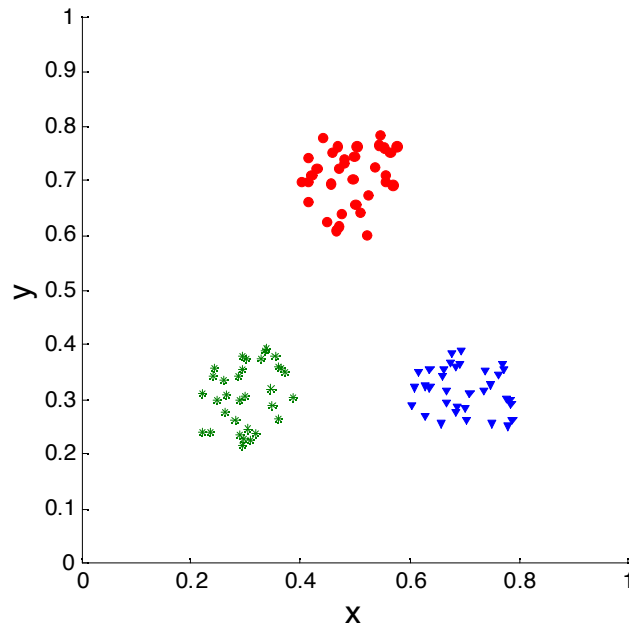
Corr = -0.9235



Corr = -0.5810

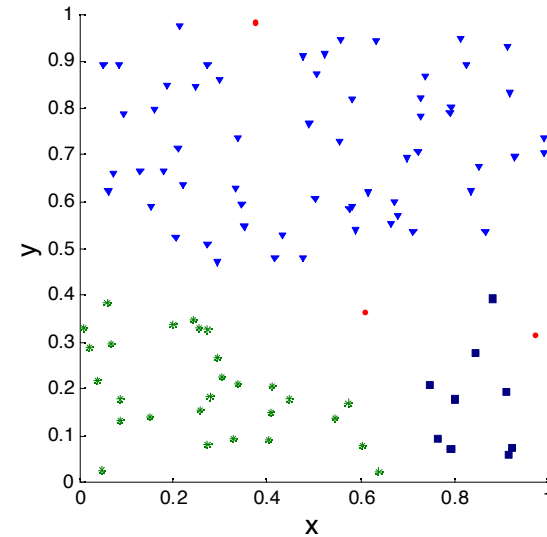
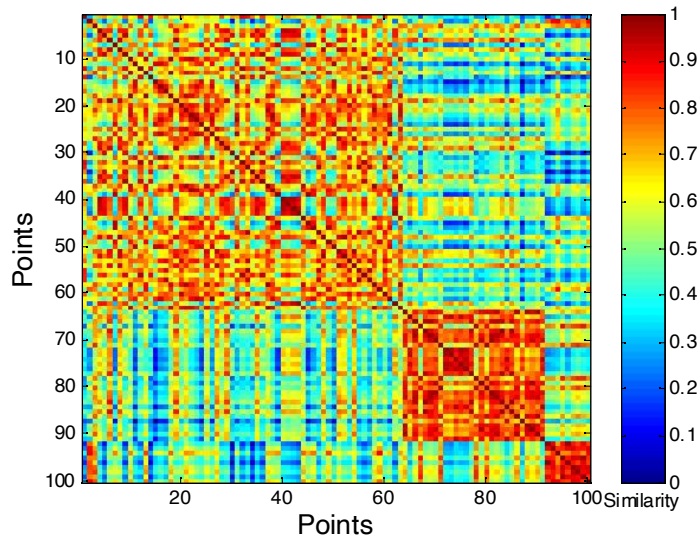
# Using Similarity Matrix for Cluster Validation

- Order the similarity matrix with respect to cluster labels and inspect visually.



# Using Similarity Matrix for Cluster Validation

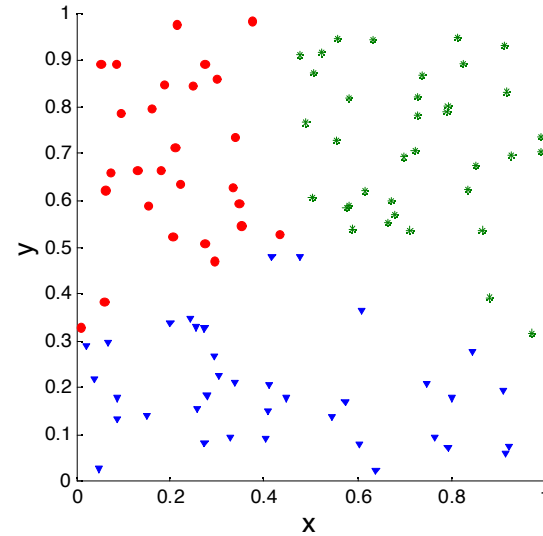
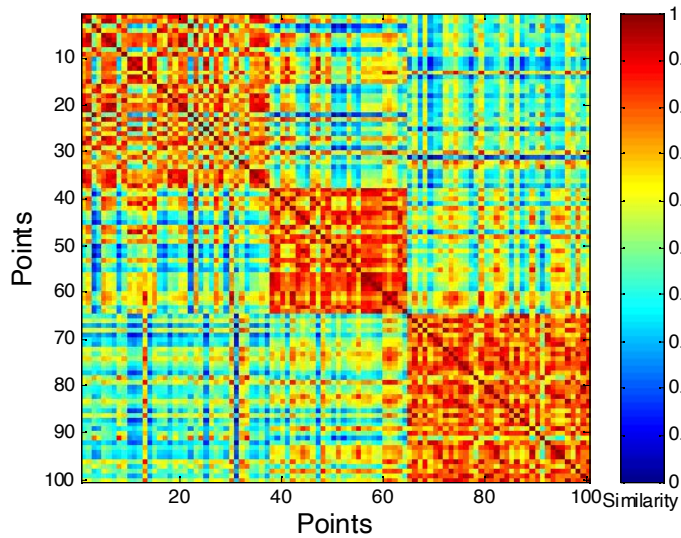
- Clusters in random data are not so crisp



DBSCAN

# Using Similarity Matrix for Cluster Validation

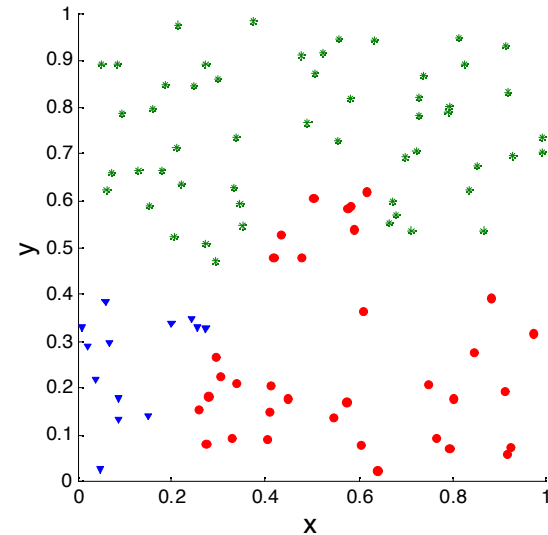
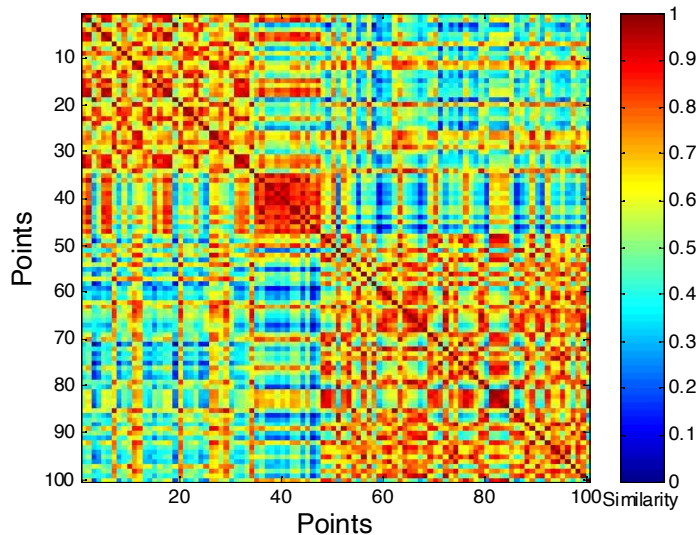
- Clusters in random data are not so crisp



K-means

# Using Similarity Matrix for Cluster Validation

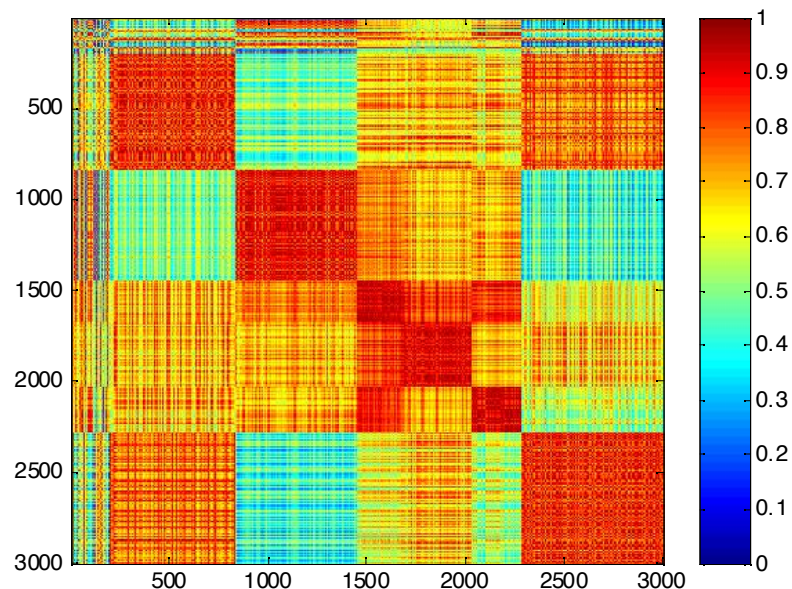
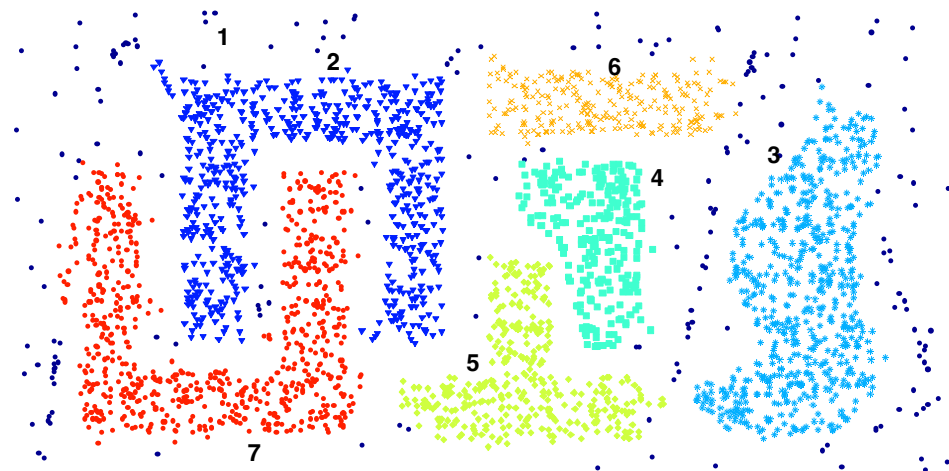
- Clusters in random data are not so crisp



Complete Link



# Using Similarity Matrix for Cluster Validation

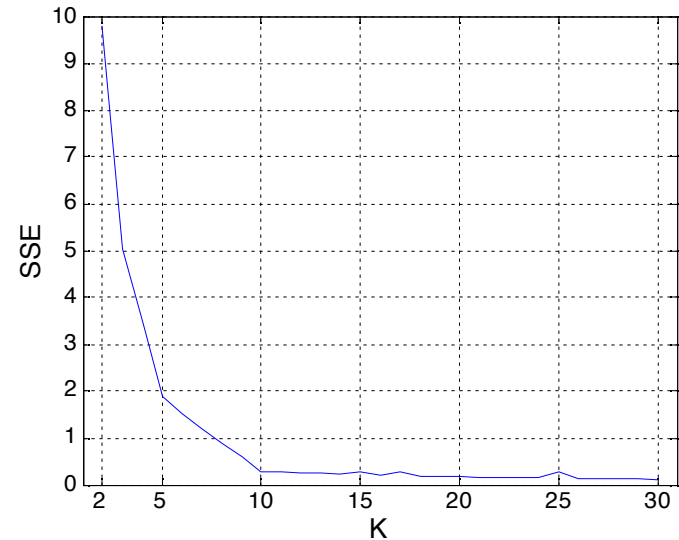
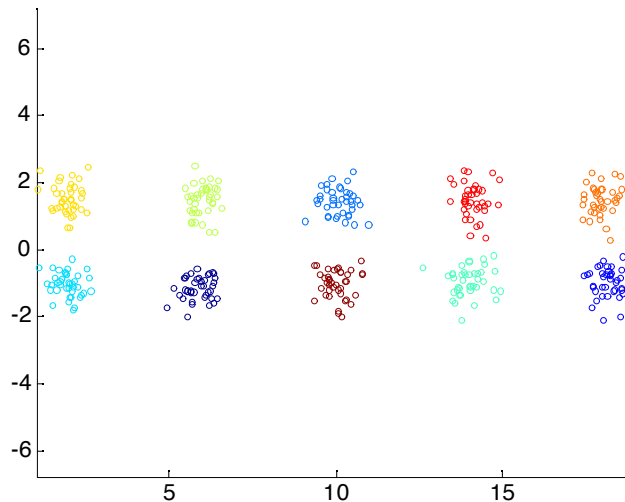


DBSCAN



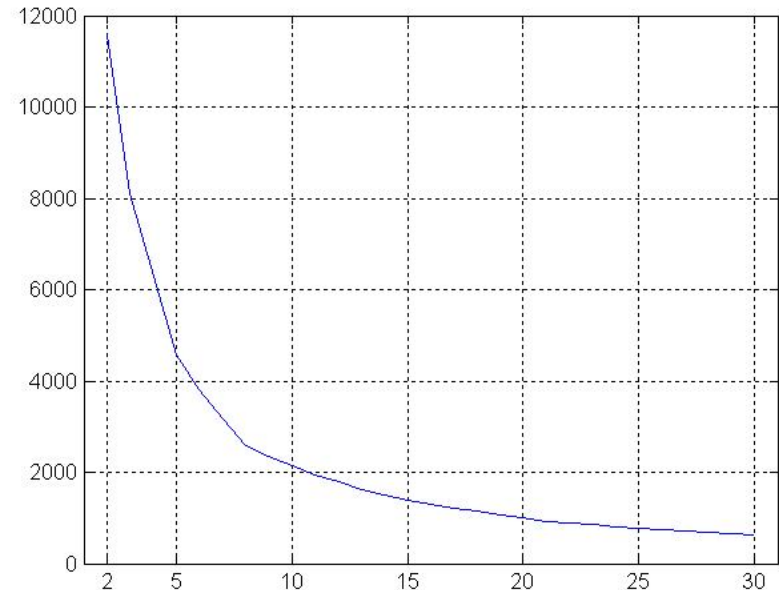
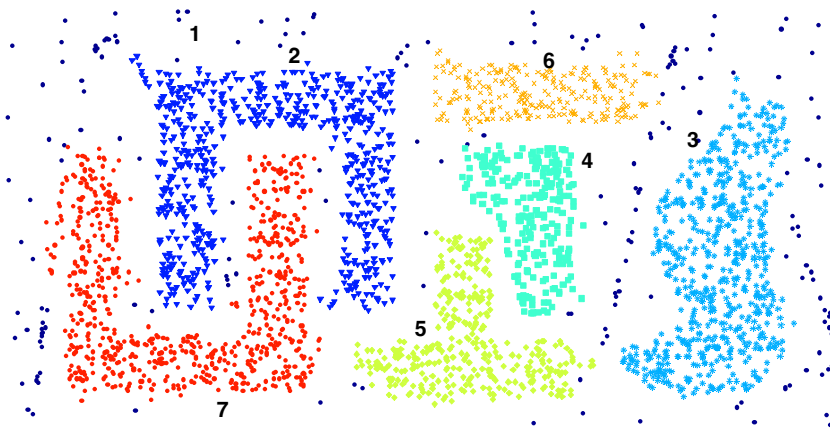
# Internal Index and # Clusters

- SSE is good for comparing two clusterings or two clusters (average SSE).
- Can also be used to estimate the number of clusters



# Internal Index and # Clusters

- SSE curve for a more complicated data set



SSE of clusters found using K-means

# External Index and Cluster Validity

- Classification-oriented measures:

| Cluster | Entertainment | Financial | Foreign | Metro | National | Sports | Entropy | Purity |
|---------|---------------|-----------|---------|-------|----------|--------|---------|--------|
| 1       | 3             | 5         | 40      | 506   | 96       | 27     | 1.2270  | 0.7474 |
| 2       | 4             | 7         | 280     | 29    | 39       | 2      | 1.1472  | 0.7756 |
| 3       | 1             | 1         | 1       | 7     | 4        | 671    | 0.1813  | 0.9796 |
| 4       | 10            | 162       | 3       | 119   | 73       | 2      | 1.7487  | 0.4390 |
| 5       | 331           | 22        | 5       | 70    | 13       | 23     | 1.3976  | 0.7134 |
| 6       | 5             | 358       | 12      | 212   | 48       | 13     | 1.5523  | 0.5525 |
| Total   | 354           | 555       | 341     | 943   | 273      | 738    | 1.1450  | 0.7203 |

**entropy** For each cluster, the class distribution of the data is calculated first, i.e., for cluster  $j$  we compute  $p_{ij}$ , the ‘probability’ that a member of cluster  $j$  belongs to class  $i$  as follows:  $p_{ij} = m_{ij}/m_j$ , where  $m_j$  is the number of values in cluster  $j$  and  $m_{ij}$  is the number of values of class  $i$  in cluster  $j$ . Then using this class distribution, the entropy of each cluster  $j$  is calculated using the standard formula  $e_j = \sum_{i=1}^L p_{ij} \log_2 p_{ij}$ , where the  $L$  is the number of classes. The total entropy for a set of clusters is calculated as the sum of the entropies of each cluster weighted by the size of each cluster, i.e.,  $e = \sum_{i=1}^K \frac{m_i}{m} e_j$ , where  $m_j$  is the size of cluster  $j$ ,  $K$  is the number of clusters, and  $m$  is the total number of data points.

**purity** Using the terminology derived for entropy, the purity of cluster  $j$ , is given by  $purity_j = \max p_{ij}$  and the overall purity of a clustering by  $purity = \sum_{i=1}^K \frac{m_i}{m} purity_j$ .

# External Index and Cluster Validity

---

- Similarity-oriented measures:
  - ideal class similarity matrix
  - ideal cluster similarity matrix
  - correlation between the two matrixes
- An example:

Clusters:  $C1=\{p1\ p2\ p3\}$ ,  $C2=\{P4\ P5\}$   
Class:  $L1=\{P1\ P2\}$ ,  $L2=\{P3\ P4\ P5\}$

# External Index and Cluster Validity

|    | P1 | P2 | P3 | P4 | P5 |
|----|----|----|----|----|----|
| P1 | 1  | 1  | 1  | 0  | 0  |
| P2 | 1  | 1  | 1  | 0  | 0  |
| P3 | 1  | 1  | 1  | 0  | 0  |
| P4 | 0  | 0  | 0  | 1  | 1  |
| P5 | 0  | 0  | 0  | 1  | 1  |

Ideal cluster similarity matrix

|    | P1 | P2 | P3 | P4 | P5 |
|----|----|----|----|----|----|
| P1 | 1  | 1  | 0  | 0  | 0  |
| P2 | 1  | 1  | 0  | 0  | 0  |
| P3 | 0  | 0  | 1  | 1  | 1  |
| P4 | 0  | 0  | 1  | 1  | 1  |
| P5 | 0  | 0  | 1  | 1  | 1  |

Ideal class similarity matrix

- Correlation between the two matrixes:
  - Rand-Sta.  $(f_{00}+f_{11})/(f_{00}+f_{11}+f_{01}+f_{10})$
  - Jaccard  $(f_{11})/(f_{11}+f_{01}+f_{10})$



# Outline

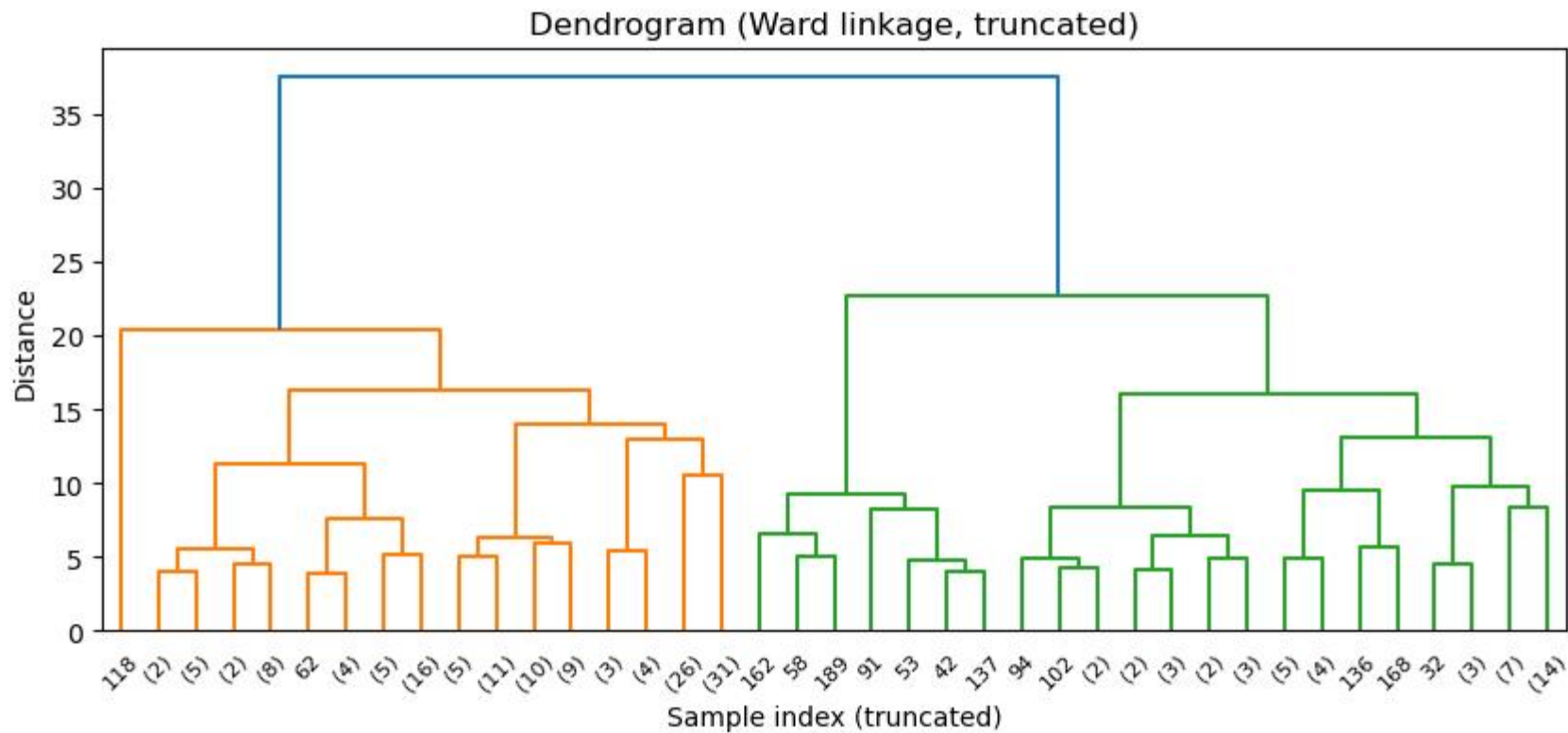
---

- Hierarchical Clustering
- DBSACN
- Cluster Validation
- Case Study

## Bank Marketing Data:

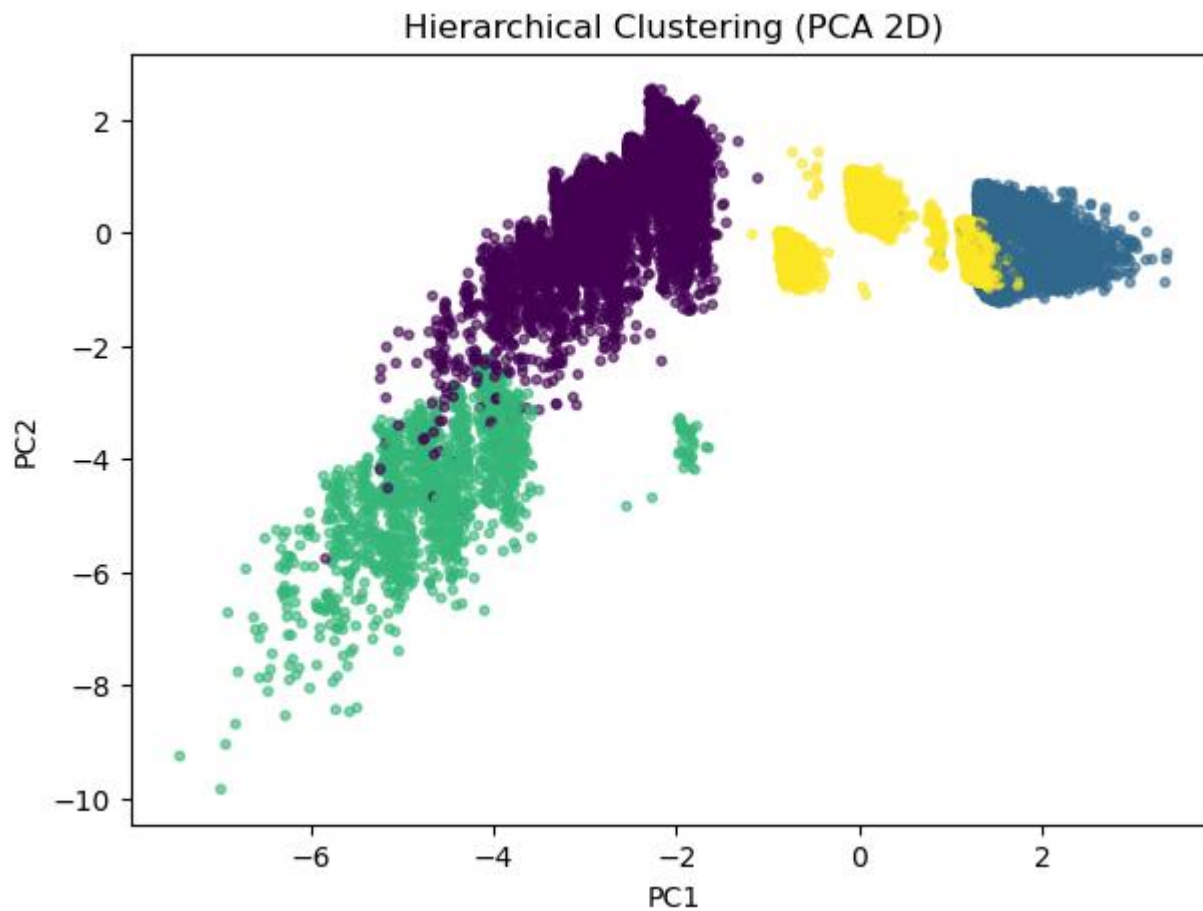
- Bank client data:
  - 1 - age (numeric)
  - 2 - job : type of job (categorical: 'admin.','blue-collar','entrepreneur','housemaid','management','retired','self-employed','services','student','technician','unemployed','unknown')
  - 3 - marital : marital status (categorical: 'divorced','married','single','unknown'; note: 'divorced' means divorced or widowed)
  - 4 - education (categorical: 'basic.4y','basic.6y','basic.9y','high.school','illiterate','professional.course','university.degree','unknown')
  - 5 - default: has credit in default? (categorical: 'no','yes','unknown')
  - 6 - housing: has housing loan? (categorical: 'no','yes','unknown')
  - 7 - loan: has personal loan? (categorical: 'no','yes','unknown')
- Related with the last contact of the current campaign:
  - 8 - contact: contact communication type (categorical: 'cellular','telephone')
  - 9 - month: last contact month of year (categorical: 'jan', 'feb', 'mar', ..., 'nov', 'dec')
  - 10 - day\_of\_week: last contact day of the week (categorical: 'mon','tue','wed','thu','fri')
  - 11 - duration: last contact duration, in seconds (numeric). Important note: this attribute highly affects the output target (e.g., if duration=0 then y='no'). Yet, the duration is not known before a call is performed. Also, after the end of the call y is obviously known. Thus, this input should only be included for benchmark purposes and should be discarded if the intention is to have a realistic predictive model.
- Other attributes:
  - 12 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
  - 13 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric; 999 means client was not previously contacted)
  - 14 - previous: number of contacts performed before this campaign and for this client (numeric)
  - 15 - poutcome: outcome of the previous marketing campaign (categorical: 'failure','nonexistent','success')
- Social and economic context attributes
  - 16 - emp.var.rate: employment variation rate - quarterly indicator (numeric)
  - 17 - cons.price.idx: consumer price index - monthly indicator (numeric)
  - 18 - cons.conf.idx: consumer confidence index - monthly indicator (numeric)
  - 19 - euribor3m: euribor 3 month rate - daily indicator (numeric)
  - 20 - nr.employed: number of employees - quarterly indicator (numeric)
- Desired target:
  - 21 - y - has the client subscribed a term deposit? (binary: 'yes','no')

# Case Study

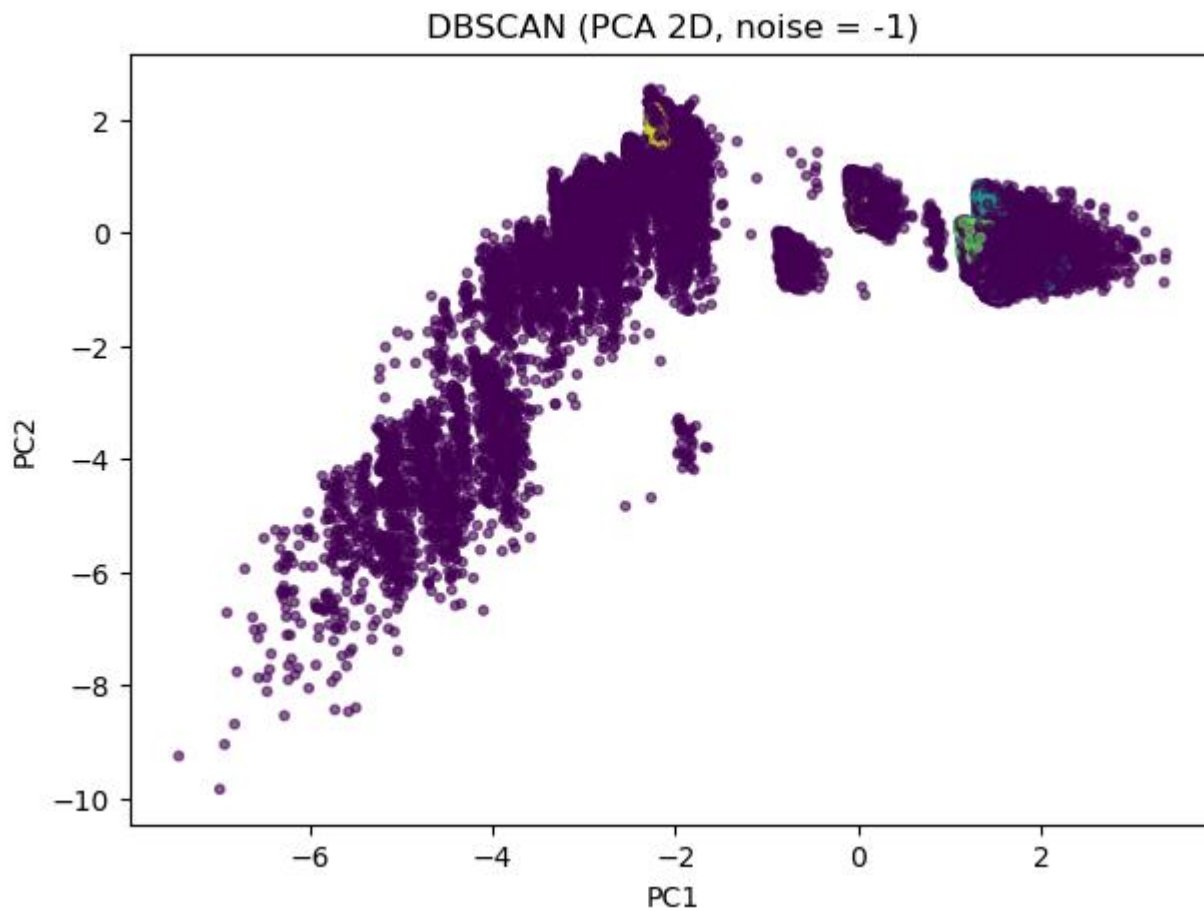




# Case Study



# Case Study





# Some Takeaways

---

- Hierarchical Clustering
- DBSACN
- Cluster Validation



# Acknowledgement

---

- Introduction to data mining (by Tan, Steinbach, Kumar)
- NYU: machine learning course (by Xi Chen)

# The End

---

■ THANK YOU

Any questions?

